

# 2018 Level I Mock Exam (C) AM

The morning session of the 2018 Level I Chartered Financial Analyst® Mock Examination has 120 questions. To best simulate the exam day experience, candidates are advised to allocate an average of one and a half minutes per question for a total of 180 minutes (3 hours) for this session of the exam.

| Questions     | Topic                              | Minutes    |
|---------------|------------------------------------|------------|
| 1–18          | Ethical and Professional Standards | 27         |
| 19–33         | Quant                              | 22.5       |
| 34–45         | Econ                               | 18         |
| 46–69         | Financial Reporting and Analysis   | 36         |
| 70–78         | Corporate Finance                  | 13.5       |
| 79–86         | Portfolio Management               | 12         |
| 87–98         | Equity                             | 18         |
| 99–110        | Fixed Income                       | 18         |
| 111–115       | Derivatives                        | 7.5        |
| 116–120       | Alternative Investments            | 7.5        |
| <b>Total:</b> |                                    | <b>180</b> |

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## 2018 LEVEL I MOCK EXAM (C) AM

- 1 Ron Dunder, CFA, is the CIO for Bling Trust (BT), an investment advisor. Dunder recently assigned one of his portfolio managers, Doug Chetch, to manage several accounts that primarily invest in thinly traded micro-cap stocks. Dunder soon notices that Chetch places many stock trades for these accounts on the last day of the month, toward the market's close. Dunder finds this trading activity unusual and speaks to Chetch who explains that the trading activity was completed at the client's request. Dunder does not investigate further. Six months later, regulatory authorities sanction BT for manipulating micro-cap stock prices at month end in order to boost account values. Did Dunder violate any CFA Institute Standards of Professional Conduct?
- A No.
  - B Yes, because he failed to reasonably supervise Chetch.
  - C Yes, because he did not report his findings to regulatory authorities.

B is correct because the CFA Institute Standard on Responsibilities of Supervisors, Standard IV(C), requires members/candidates to take steps to detect and prevent violations of laws, rules, and regulations. Dunder failed in his supervisory role when he accepted Chetch's explanation of the unusual trading activity. Dunder should have reviewed the client's goals and objectives and records to see if they in fact requested month-end trading. Regardless of the explanation provided by Chetch, Dunder should have investigated further.

A is incorrect because there was a violation of the Standards.

C is incorrect because the Standards do not require that the findings be reported to regulatory authorities.

Guidance for Standards I–VII

LOS a

Standard IV(C)–Responsibilities of Supervisors

- 2 David Donnigan enrolled to take the Level II CFA examination in the current year, however he did not take the exam. Donnigan advised his employer he passed Level II. Subsequently, he registered to take the Level II exam the next year. Which CFA Institute Standard of Professional Conduct did Donnigan *least likely* violate?
- A Duty to employer
  - B Professional misconduct
  - C Referencing Candidacy in the CFA Program

C is correct because as he registered to take the exam in the next year he still qualifies to state he is a candidate in the CFA Program. He would not, however, be authorized to reference that he is a Level III candidate and if asked would need to specify that he is a Level II candidate.

A is incorrect because Standard IV(A) requires members and candidates to protect the interests of their firm by refraining from any conduct that would injure the firm, deprive it of profit, or deprive it of the member's or candidate's skills and ability. Lying about passing the CFA examination could cause the firm to misrepresent Donnigan's progression in the CFA program to a client, leading to reputational damage if discovered.

B is incorrect as he lied to his employer about passing the examination, which is a professional misconduct violation. Members should not engage in conduct involving dishonesty, fraud, deceit, or misrepresentation [Standard I(D)].

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Guidance for Standards I–VII

LOS a

Standard I(D)–Misconduct, Standard IV(A)–Loyalty

- 3 Delaney O'Keefe, a CFA candidate, is a portfolio manager at Bahati Management Company. The company is considering investing offshore for the first time, particularly in North America, on behalf of their clientele, all of whom are high-net-worth individuals. O'Keefe does not have experience in offshore investments, so she hires Mark Carlson, CFA, of Carlson Consulting on the sole basis that he has a CFA charter, to undertake due diligence exercises on the top ten portfolio managers in North America, ranked by Assets under Management (AUM). To avoid violating any Code and Standards, O'Keefe should *most likely* undertake:

- A a sampling of the suitability of North America for clients.
- B the due diligence exercise on the top ten asset managers herself.
- C a due diligence exercise on Mark Carlson and Carlson Consulting.



C is correct because O'Keefe can delegate a due diligence exercise to a third party but must ensure the person or company hired to do so is competent with the skills necessary to undertake a thorough and appropriate analysis. While Carlson may be qualified to undertake this assignment, O'Keefe needs to take the necessary steps to ensure that this is the case. Just because a person has a CFA charter does not necessarily mean they are appropriate for the assignment.

A is incorrect. O'Keefe should undertake an analysis of all of the clients in terms of suitability, not just a sampling.

B is incorrect because O'Keefe can delegate a due diligence exercise to a third party as long as she is diligent in assessing who should be appointed to undertake the assignment.

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Guidance for Standards I–VII

Standard V(A)–Diligence and Reasonable Basis

LOS a, c

- 4 Sanctions imposed by CFA Institute for violations of the CFA Institute Code of Ethics or Standards of Professional Conduct *least* likely include:
- A monetary fines.
  - B public censure.
  - C revocation of a CFA Charter.



A is correct. Sanctions available to CFA Institute do not include monetary fines. However, sanctions imposed by CFA Institute may have significant consequences; they include public censure, suspension of membership and use of the CFA designation, and revocation of the CFA charter. Candidates enrolled in the CFA Program who have violated the Code and Standards or testing policies may be suspended or prohibited from further participation in the CFA Program.

B is incorrect. Sanctions imposed by CFA Institute may have significant consequences; they do include public censure.

C is incorrect. Sanctions imposed by CFA Institute may have significant consequences; they do include revocation of the CFA charter.

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Code of Ethics and Standards of Professional Conduct  
LOS a

5 Verification of compliance with the GIPS standards *most likely* requires:

- A an independent third party to carry out the verification.
- B an assurance that the composite presentations are accurate.
- C verification for each specific composite under review.



A is correct. While a firm is responsible for their own compliance claim, they cannot perform their own verification. An independent third party must undertake the verification.

B is incorrect because verification does not ensure the accuracy of any specific composite presentation.

C is incorrect because verification is performed with respect to an entire firm, not on specific composites.

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Introduction to the Global Investment Performance Standards (GIPS)  
LOS a  
Section V

6 According the GIPS standards, for periods beginning on or after 1 January 2011, the aggregate fair value of total firm assets *most likely* includes all:

- A fee-paying discretionary accounts.
- B fee- and non-fee-paying discretionary accounts.
- C fee- and non-fee-paying discretionary and non-discretionary accounts.



C is correct because for periods beginning on or after 1 January 2011, total firm assets must include the aggregate fair value of all discretionary and non-discretionary assets managed by the firm. This includes both fee-paying and non-fee-paying portfolios (0.A.13).

A is incorrect because with effect from 1 January 2011, Total Firm Assets include the aggregation of the fair value of all discretionary accounts, not just fee-paying discretionary accounts. It also includes non-discretionary accounts.

B is incorrect because with effect from 1 January 2011, Total Firm Assets include the fair value of non-discretionary accounts as well as discretionary accounts.

Global Investment Performance Standards (GIPS)

LOS a

GIPS Requirement 0.A.13

- 7 In countries where new local laws relating to calculation and presentation of investment performance conflict with GIPS standards, firms who have claimed GIPS compliance should *most likely*:
- A stop claiming GIPS compliance.
  - B follow local laws, continue to claim GIPS compliance, and disclose conflicts.
  - C continue to claim GIPS compliance, disclosing non-compliance with new laws.

B is correct because where local laws and regulations regarding calculation and presentation conflict with GIPS standards, firms must abide by the local laws and regulations. They are still allowed to claim GIPS compliance but must disclose areas where the local requirements conflict with those of the GIPS standards.

A is incorrect because while firms must abide by local requirements firms are still allowed to claim GIPS compliance but must disclose areas where the local requirements conflict with those of the GIPS standards.

C is incorrect because while firms must abide by local requirements firms are still allowed to claim GIPS compliance but must disclose areas where the local requirements conflict with those of the GIPS standards.

Global Investment Performance Standards (GIPS)

LOS c

- 8 James Simone, CFA, the CFO of a publicly listed company, seeks to improve the quality of his company's communication with institutional fund managers. He holds an investor briefing with this group the evening before the company earnings are announced. The company's quarterly earnings are broadcast in a press release the next day before the market opens. The earnings information in the investor briefing is identical to that in the press release. Did Simone *most likely* violate the CFA Institute Standards of Professional Conduct?
- A Yes
  - B No, because investor briefing and press release information are identical.
  - C No, because the company releases information while the market is closed.

A is correct because Simone violated Standard II(A)—Material Nonpublic Information by giving institutional fund managers access to material nonpublic information prior to public dissemination (i.e., the press release). By releasing earnings results to a select group of institutional fund managers prior to a public press release, Simone allows the institutional fund managers a time advantage over other investors not invited to the investor briefing.

B is incorrect because it is the timing of the release of the information that causes a violation of Standard II(A).

C is incorrect because despite releasing the information after the close of the market, institutional investors have been given access to nonpublic information in advance of any public dissemination through a press release in violation of Standard II(A).

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Guidance for Standards I–VII

LOS b

Standard II(A)–Material Nonpublic Information

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- 9 Johannes Meir, CFA, is a compliance officer for Family Estate Planning, LLC, a private wealth consulting firm. Many of his colleagues have family members who have started their own retail businesses. Some of Meir's colleagues have been asked by relatives to serve as non-executive directors or advisors to their companies. Meir should *most likely* recommend which of the following policies to ensure compliance with the CFA Institute Standards?
- A Prohibit employees from becoming directors or advisors
  - B Require employees to declare all income sources annually
  - C Require employees to declare all outside business interests



C is correct because Standard VI(A) requires the disclosure of conflicts. For Meir to understand what potential conflicts of interest employees may have with the firm and with their clients, he would need to know the outside interests of each staff member. The staff members themselves may not know enough about the company and its clients to disclose those interests that would present a potential conflict. Therefore, it may be best to have all employees declare their outside business interests on an annual basis so Meir can make the determination as to what outside business interests need to be disclosed to clients.

A is incorrect because the Standards do not prohibit employees from becoming directors, advisors, or having other outside interest. The concern is to ensure that employees' activities do not create conflicts or potential conflicts of interests with their clients and or firms.

B is incorrect because the Standards do not require the reporting of all income, only referral fees [Standard VI(C)] and additional compensation [Standard IV(B)] that would be sourced from activities that would likely compete with the employer. Even if the employees do not know the client list of the company they could likely determine if any income they receive is in competition with their firm's as they would know the general sources of the firm's income.

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Guidance for Standards I–VII

LOS c

Standard VI(A)–Disclosure of Conflicts

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- 10 Robin Herring, CFA, is a government bond research analyst at an independent credit rating agency. A competitor credit rating agency just downgraded the bonds of a government Herring follows. Herring notes that all of the information in the competitor's report was covered in his analysis published last week. In the past, Herring has been slow to downgrade bonds, so he starts to doubt his own analysis after seeing the competitor's report. Herring decides to reissue his credit rating of this government bond and match the competitor's downgrade. In his revised report, Herring states that new information has been made available to justify the downgrade. Herring posts the revision on the credit

rating agency's website and provides it by e-mail to all clients who received the original. Herring's rating change least likely violated which of the following CFA Institute Code of Ethics and Standards of Professional Conduct?

- A** Fair Dealing
- B** Communication with Clients
- C** Diligence and Reasonable Basis

A is correct because the analyst has dealt fairly with all clients by sending them an e-mail and posting his rating change on the credit rating agency's website when making material changes to his prior investment recommendation; therefore, he has not violated Standard III(B)–Fair Dealing. Clients should be treated fairly when material changes in a member's or candidate's prior investment recommendations are disseminated, which has been done.

B is incorrect because the analyst has used the release of a competitor's report, contrary to his own previously published report, as a reason to revise his recommendation and has not used reasonable judgment in identifying which factors are important to his investment analyses, recommendations, or actions, as required by Standard V(B)–Communication with Clients and Prospective Clients.

C is incorrect because the analyst does not have a reasonable or adequate basis for his downgrade, as required by Standard V(A)–Diligence and Reasonable Basis. The analyst has also violated Standard I(C)–Misrepresentation which prohibits making misrepresentations relating to investment analysis, recommendations, actions, or other professional activities.

Guidance for Standards I–VII

LOS b

Standard I(C)–Misrepresentation, Standard III(B)–Fair Dealing, Standard V(A)–Diligence and Reasonable Basis, Standard V(B)–Communication with Clients and Prospective Clients

- 11** Carolina Ochoa, CFA, is the chief financial officer at Pantagonia Computing. Ochoa is currently the subject of an inquiry by Pantagonia's corporate investigations department. The inquiry is the result of an anonymous complaint accusing Ochoa of falsifying travel expenses for senior management related to a government contract. According to the CFA Institute Code of Ethics and Standards of Professional Conduct, it is most appropriate for Ochoa to disclose the allegations:

- A** on her Professional Conduct Statement.
- B** to CFA Institute when the investigation concludes.
- C** to CFA Institute if the allegations are proven correct.

A is correct because members and candidates must self-disclose on the annual Professional Conduct Statement all matters that question their professional conduct, such as involvement in civil litigation or criminal investigations or being the subject of a written complaint.

B is incorrect because members and candidates must self-disclose on the annual Professional Conduct Statement all matters that question their professional conduct such as involvement in civil litigation or criminal investigation or being the subject of a written complaint.

C is incorrect because members and candidates must self-disclose on the annual Professional Conduct Statement all matters that question their professional conduct such as involvement in civil litigation or criminal investigation or being the subject of a written complaint.

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Code of Ethics and Standards of Professional Conduct

LOS c

Section 1.1

- 12** Jan Loots, CFA, quit his job as a portfolio manager at an investment firm with whom he had a non-solicitation agreement he signed several years ago. Loots received permission to take his investment performance history with him and also took a copy of the firm's software-trading platform. Subsequently, Loots sent out messages on social media sites announcing he was looking for clients for his new investment management firm. Access to Loots' social media sites is restricted to friends, family, and former clients. Loots least likely violated the CFA Institute Standards of Professional Conduct concerning his:

- A** trading software.
- B** non-solicitation agreement.
- C** investment performance history.



C is correct because the portfolio manager received permission to use his investment performance history from his prior employer. The member violated his non-solicitation agreement by indicating his availability to new clients on several social media sites accessible by clients of his former employer. This is a violation of Standard IV(A)–Loyalty because he did not act for the benefit of his former employer. In this case, the member may cause harm to his former employer if his weekend messages result in clients moving to his new business from his former employer. The member also violated this standard by taking his employer's property, trading software.

A is incorrect because the portfolio manager took property of his former employer, proprietary trading software, and violated Standard IV(A)–Loyalty. Although the manager created the software himself, it was during a period of time when the large money manager employed him and the software is not his property to take.

B is incorrect because the member violated Standard IV(A)–Loyalty as he did not act for the benefit of his former employer. In this case, the member may cause harm to his former employer if his weekend messages result in clients moving to his new business.

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Guidance for Standards I–VII

LOS c

Standard IV(A)–Loyalty

- 13** Gabrielle Gabbe, CFA, has been accused of professional misconduct by one of her competitors. The allegations concern Gabbe's personal bankruptcy filing ten years ago when she was a college student and had a large amount of medical bills she could not pay. By not disclosing the bankruptcy filing to her clients, did Gabbe *most likely* violate any CFA Institute Standards of Professional Conduct?
- A** No.
  - B** Yes, related to Misconduct.
  - C** Yes, related to Misrepresentation.





A is correct as a personal bankruptcy does not necessarily constitute a violation of Standard I(D). If the circumstances of the bankruptcy involved fraudulent or deceitful business conduct, then failing to disclose it may constitute a violation of the Standards.

B is incorrect because there has not been a violation of the Standards.

C is incorrect because there has not been a violation of the Standards.

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Guidance for Standards I–VII  
LOS b

- 14** Bailey Watson, CFA, manages 25 emerging market pension funds. He recently had the opportunity to buy 100,000 shares in a publicly listed company whose prospects are considered “above industry norm” by most analysts. The company’s shares rarely trade because most managers take a “buy and hold” strategy due to the company’s small free float. Before placing the order with his dealer, Watson allocated the shares to be purchased according to the weighted value of each of his clients’ portfolios. When it came time to execute the trades, the dealer was only able to purchase 50,000 shares. To prevent violating Standard III(B)–Fair Dealing, it would be *most* appropriate for Watson to reallocate the 50,000 shares purchased by:

- A** reducing each pension fund’s allocation proportionately.
- B** distributing them equally amongst all the pension fund portfolios.
- C** allocating randomly but giving funds left out priority on the next similar type trade.



A is correct because Standard III(B)–Fair Dealing requires members and candidates to deal fairly and objectively with all clients. Certain clients cannot be favored over other clients when their investment objectives and circumstances are similar. Therefore, the most appropriate way to handle the reallocation of an illiquid share is to reduce each client’s proportion on a pro rata, or weighted basis.

B is incorrect because it may not be appropriate to distribute the purchased shares on an equal basis, as the client funds may be dissimilar in size, which would result in the smaller funds being given preference over the larger funds.

C is incorrect due to the shares being highly illiquid so it is not certain when the next opportunity will arise for the shares to be purchased again, thus discriminating against the clients who were left out of the initial allocation. In cases where the shares are liquid thus trades are certain, this may be a proper method to handle the initial under allocation.

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Guidance for Standards I–VII  
LOS c

- 15** Jorge Lopez, CFA, is responsible for proxy voting on behalf of his bank’s asset management clients. Lopez recently performed a cost–benefit analysis, showing that proxy voting analysis might not benefit the bank’s clients. As a result, Lopez immediately changes the proxy voting policies and procedures without informing anyone else of the change. Lopez now votes client proxies on the side of management on all issues with the exception of major mergers where a significant impact on the stock price is expected. Lopez *least likely* violated the CFA Institute Standards of Professional with regards to:

- A** cost–benefit analysis.

- B** voting with management.
- C** proxy voting policy disclosures.

A is correct because there is no violation of Standard III(A)–Loyalty, Prudence, and Care by performing a cost–benefit analysis showing that voting all proxies might not benefit the client, and concluding voting proxies may not be necessary in all instances. However, even though voting proxies may not be necessary in all instances, part of a member’s or candidate’s duty of loyalty under Standard III(A)–Loyalty, Prudence, and Care, includes voting proxies in an informed and responsible manner, which is not being done by Lopez by automatically voting with management on the majority of issues. In addition, members and candidates should disclose to clients their proxy voting policies, including any changes to that policy as required by Standard III(A)–Loyalty, Prudence, and Care, which has not been done.

B is incorrect because even though voting proxies may not be necessary in all instances, part of a member’s or candidate’s duty of loyalty under Standard III(A)–Loyalty, Prudence, and Care, includes voting proxies in an informed and responsible manner, which is not being done by Lopez.

C is incorrect because members and candidates should disclose to clients their proxy voting policies, including any changes to that policy as required by Standard III(A)–Loyalty, Prudence, and Care.

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Guidance for Standards I–VII

LOS b

Standard III(A)–Loyalty, Prudence, and Care

- 16** Sue Kim, CFA, a US citizen, works as an analyst for a subsidiary of a US investment firm on a small island that attracts offshore investment accounts. Local securities laws allow insider trading. While having dinner with the CEO of a local company, Kim learns that the firm is in negotiations to be acquired for a significant premium. Would Kim *most likely* comply with the CFA Institute Standards if she purchased the company’s shares for her client accounts?
- A** No.
  - B** Yes, local laws allow insider trading.
  - C** Yes, if she receives permission from compliance department.

A is correct because Standard II(A) prohibits members or candidates from acting on material nonpublic information no matter if local laws or a compliance department allow it. In the event of conflict between the Code and Standards and local laws, Standard I(A) requires members or candidates to comply with the stricter law, rule, or regulation.

B is incorrect because although the local securities laws allow for insider trading, members and candidates are required to follow the stricter of local law or the Code and Standards. In this case, the Code and Standards are stricter than local law and need to be followed by Kim.

C is incorrect because even if her compliance department would approve this transaction, it is still a violation of the Codes and Standards and should not be acted on by Kim.

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Guidance for Standards I–VII

LOS b

Standard I(A)–Knowledge of the Law, Standard II(A)–Material Nonpublic Information

- 17 Preeta Singh, a CFA candidate, is an asset manager employed by a fund management company managing very large segregated pension funds. In her spare time outside of working hours, Singh likes to provide management consulting services to small companies to help grow their businesses, focusing on strategic planning. Singh is paid for the consulting services and has also provided her employer information about these outside activities. Does Singh *most likely* violate the CFA Code of Ethics with regard to Duties to Employers?
- A No.
  - B Yes, with regard to Loyalty.
  - C Yes, with regard to Additional Compensation Arrangements.

A is correct because Singh does not violate any Standard relating to Duties to Employers. She conducts unrelated non-competitive services to clients outside of business hours and thus does not deprive her employer of the advantage of her skills and abilities, nor is there any indication that she divulges confidential information or otherwise causes harm to her employer. She also does not need to divulge information relating to her additional compensation or to seek permission, as her management consulting services do not create a conflict of interest with her employer's interest.

B is incorrect because Singh conducts unrelated non-competitive services to clients outside of business hours and thus does not deprive her employer of the advantage of her skills and abilities, nor is there any indication that she divulges confidential information or otherwise causes harm to her employer.

C is incorrect because she does not need to divulge information relating to her additional compensation or to seek permission, as her management consulting services do not create a conflict of interest with her employer's interest.

Guidance for Standards I–VII

LOS b

Standard IV(A)–Loyalty, Standard IV(B)–Additional Compensation Arrangements

- 18 PNW Bank publishes *Investment Monthly* magazine, which highlights a specific stock in each issue. Publication of the magazine invariably causes the high-lighted stocks to rise significantly in value. Rachel Coursing, CFA, manager of PNW's marketing department, often trades in the securities mentioned in the *Investment Monthly* articles prior to publication of the magazine. Coursing has access to the recommendations prior to the magazine's publication because the magazine is created in her department and edited by her. PNW's Code of Ethics restricts trading by all of the bank's analysts and portfolio managers and requires their trades to be pre-cleared by the Compliance Department. Coursing *least likely* violated which of the following CFA Institute Standards of Professional Conduct?
- A Priority of Transactions
  - B Diligence and Reasonable Basis
  - C Material Nonpublic Information

B is correct because Coursing has not violated Standard V–Investment Analysis, Recommendations, and Actions, as she is not analyzing investments, making investment recommendations, or taking investment actions for clients. Coursing has violated Standard VI(B)–Priority of Transactions as clients of the bank have not been given priority

over investment transactions in which a member or candidate is the beneficial owner. In addition, Coursing violated Standard II(A)–Material Nonpublic Information by trading on material nonpublic information. The *Investment Monthly* article written by PNW is considered nonpublic until the magazine is widely distributed, and publication of the magazine will materially impact the market price of stocks highlighted. Even though Coursing is not required by her bank to pre-clear her trades, she is restricted from trading by Standard II(A).

A is incorrect because Coursing has violated Standard VI(B)–Priority of Transactions as clients have not been given priority over investment transactions in which a member or candidate is the beneficial owner.

C is incorrect because Coursing violated Standard II(A)–Material Nonpublic Information by trading on material nonpublic information. The *Investment Monthly* article written by PNW is considered nonpublic until the magazine is widely distributed, and publication of the magazine will materially impact the market price of stocks highlighted. Even though Coursing is not required by her bank to pre-clear her trades, she is restricted from trading by Standard II(A).

Guidance for Standards I–VII

LOS a

Standard II(A)–Material Nonpublic Information, Standard VI(B)–Priority of Transactions

- 19 The probability of Event *A* is 40%. The probability of Event *B* is 60%. The joint probability of *AB* is 40%. The probability (*P*) that *A* or *B* occurs, or both occur, is *closest* to:

- A 40%.
- B 84%.
- C 60%.

C is correct.  $P(A \text{ or } B) = P(A) + P(B) - P(AB) = 0.40 + 0.60 - 0.40 = 0.60$  or 60%.

A is incorrect because it uses only the joint probability of 40%.

B is incorrect; it is obtained by adding 0.60 and 0.24 ( $0.60 \times 0.40$ ) = 0.84.

Probability Concepts

LOS f

Section 2

- 20 The following information applies to a portfolio composed of Fund *A* and Fund *B*:

|                                                      | Fund A | Fund B |
|------------------------------------------------------|--------|--------|
| Portfolio weights (%)                                | 70     | 30     |
| Expected returns (%)                                 | 10     | 16     |
| Standard deviations (%)                              | 7      | 13     |
| Correlation between the returns of Fund A and Fund B | 0.80   |        |

The portfolio's standard deviation of return is *closest* to:

- A 7.38%.
- B 8.80%.
- C 8.35%.

C is correct. The covariance between Fund A and Fund B, given the standard deviation of returns and the correlation between the two funds, is calculated as:

$$\text{Cov}(R_A, R_B) = \rho(R_A, R_B) \sigma(R_A) \sigma(R_B) = 0.80 \times 7\% \times 13\% = 0.00728.$$

where

$\sigma(R_A)$  and  $\sigma(R_B)$  = the standard deviations of returns of funds A and B, respectively

$\rho(R_A, R_B)$  = the correlation between the returns of funds A and B

Then the portfolio standard deviation of returns is calculated as follow:

$$\sigma(R_{\text{portfolio}}) = \left[ w_A^2 \sigma^2(R_A) + w_B^2 \sigma^2(R_B) + 2w_A w_B \text{Cov}(R_A, R_B) \right]^{0.5}$$

where  $w_A$  and  $w_B$  are the weights of funds A and B in the portfolio.

$$\begin{aligned} \sigma(R_{\text{portfolio}}) &= [0.70^2 \times 0.07^2 + 0.30^2 \times 0.13^2 + 2 \times 0.70 \times 0.30 \times 0.00728]^{0.5} \\ &= \mathbf{8.35\%} \end{aligned}$$

Alternatively, correlation is used directly in the formula for portfolio standard deviation:

$$\sigma(R_{\text{portfolio}}) = \left[ w_A^2 \sigma^2(R_A) + w_B^2 \sigma^2(R_B) + 2w_A w_B \text{Cov}(R_A, R_B) \sigma(R_A) \sigma(R_B) \right]^{0.5}$$

$$\sigma(R_{\text{portfolio}}) = [0.70^2 \times 0.07^2 + 0.30^2 \times 0.13^2 + 2 \times 0.70 \times 0.30 \times 0.80 \times 0.07 \times 0.13]^{0.5} = \mathbf{8.35\%}$$

A is incorrect. In the formula of the portfolio standard deviation of returns the term  $w_A w_B \text{Cov}(R_A, R_B)$  is used only once:

$$\begin{aligned} &\left[ w_A^2 \sigma^2(R_A) + w_B^2 \sigma^2(R_B) + w_A w_B \text{Cov}(R_A, R_B) \right]^{0.5} \\ &= [0.70^2 \times 0.07^2 + 0.30^2 \times 0.13^2 + 0.70 \times 0.30 \times 0.00728]^{0.5} = 7.38\% \end{aligned}$$

B is incorrect. It uses a wrong formula:  $w_A \sigma(R_A) + w_B \sigma(R_B) = 0.70 \times 0.07 + 0.30 \times 0.13 = 8.80\%$

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Probability Concepts

LOS k, l

Section 3

21 When rolling two six-sided dice and summing their outcomes, which of the following sums is *most likely* to occur?

- A Nine
- B Five
- C Six

C is correct. This scenario provides an example of a discrete random variable. The paired outcomes for the dice are indicated in the following table. The outcome of the dice summing to six is the most likely to occur of the three choices because it can occur in five different ways, whereas the summation to five and nine can occur in only four different ways.

| Summed Outcome | Paired Outcomes (Die 1, Die 2)             | Possible Combinations |
|----------------|--------------------------------------------|-----------------------|
| 5              | (1, 4), (2, 3), (3, 2), and (4, 1)         | 4                     |
| 6              | (1, 5), (2, 4), (3, 3), (4, 2), and (5, 1) | 5                     |
| 9              | (3, 6), (4, 5), (5, 4), and (6, 3)         | 4                     |

B is incorrect. The dice summing to five can occur in four ways, which is less than the number of ways for the dice summing to six.

A is incorrect. The dice summing to nine can occur in four ways, which is less than the number of ways for the dice summing to six.

Common Probability Distributions  
LOS b  
Section 2

22 The following information is available for a portfolio:

| Asset Class          | Asset Allocation Weight (%) | Asset Class Return (%) | Correlation with Equities Class (%) |
|----------------------|-----------------------------|------------------------|-------------------------------------|
| Equities             | 45                          | 16                     | 100                                 |
| Mortgages            | 25                          | 12                     | 30                                  |
| Cash and equivalents | 30                          | 2                      | 10                                  |

The return on the portfolio is *closest* to:

- A 10.0%.
- B 8.2%.
- C 10.8%.

C is correct. The portfolio return is the weighted mean return and is calculated as:

$$\begin{aligned}\bar{X}_w &= \sum_{i=1}^n w_i X_i \\ &= 0.45 \times 16 + 0.25 \times 12 + 0.30 \times 2 \\ &= \mathbf{10.80}.\end{aligned}$$

B is incorrect. It weights each component by the correlation with equities:  $0.45 \times 16 \times 1.00 + 0.25 \times 12 \times 0.30 + 0.30 \times 2 \times 0.10 = 8.16$ .

A is incorrect. It is the arithmetic average of the three returns without weights:  $(16 + 12 + 2)/3 = 10$ .

Statistical Concepts and Market Returns  
Section 5.4.1  
LOS e

23 The following information is available on three portfolios:

| Portfolio | Mean Return on the Portfolio (%) | Standard Deviation of the Return on the Portfolio (%) |
|-----------|----------------------------------|-------------------------------------------------------|
| D         | 10                               | 20                                                    |
| E         | 18                               | 15                                                    |
| F         | 6                                | 3                                                     |

The risk-free rate is 4%. The portfolio that has the *best* risk-adjusted performance as measured by the Sharpe ratio is:

- A Portfolio D.
- B Portfolio E.
- C Portfolio F.

B is correct.

$$S_D = (10 - 4)/20 = 0.30$$

$$S_E = (18 - 4)/15 = 0.9333$$

$$S_F = (6 - 4)/3 = 0.6667$$

The portfolio with the best risk-adjusted performance as measured by the Sharpe ratio is the one with the highest Sharpe ratio: Portfolio E.

A is incorrect. There are numerous possible mistakes that would lead a candidate to choose Portfolio D. Foremost among them would be putting the standard deviations in the numerators rather than the denominators of the Sharpe ratios. The coefficient of variation is an example: Portfolio D:  $20/10 = 2$ ; Portfolio E:  $15/18 = 0.83$ ; and Portfolio F:  $3/6 = 0.5$ .

C is incorrect. There are numerous possible mistakes that would lead a candidate to choose Portfolio F. Foremost among them would be failing to subtract the risk-free rate from the mean return of the portfolios. Portfolio D:  $10/20 = 0.5$ ; Portfolio E:  $18/15 = 1.2$ ; Portfolio F:  $6/3 = 2$ .

Statistical Concepts and Market Returns  
Section 7.8  
LOS i

- 24 In setting the confidence interval for the population mean of a normal or approximately normal distribution, and given that the sample size is small, Student's *t*-distribution is the *most* appropriate approach when the variance is:

- A known.
- B large.
- C unknown.

C is correct. When the sample size is small (and the population is normally or approximately normally distributed), the Student's *t*-distribution is preferred if the variance is unknown.

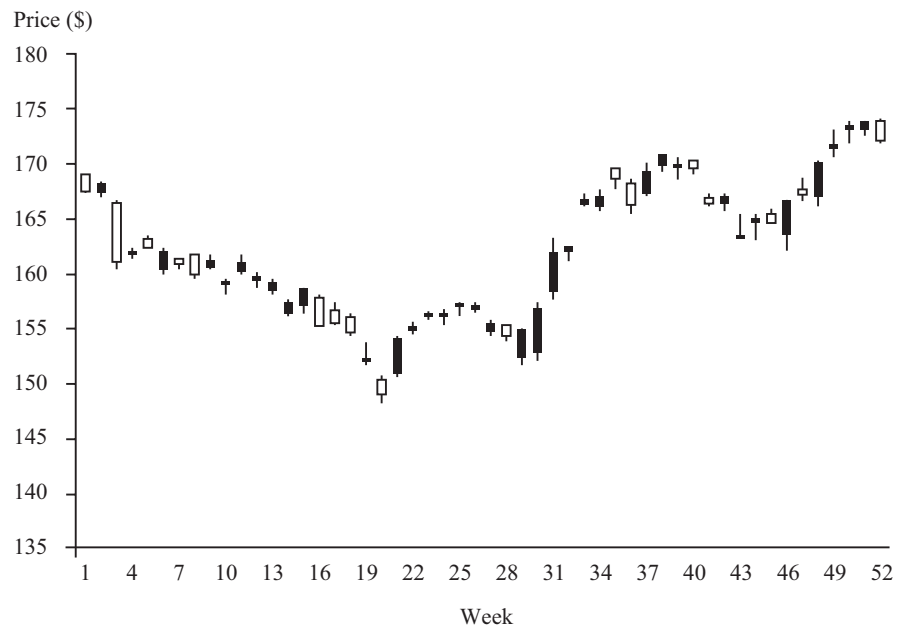
A is incorrect. The *z*-distribution is preferred if the variance is known.

B is incorrect. When there is a large sample size, not variance, it is possible to use either the z-distribution or t-distribution.

---

Sampling and Estimation  
LOS i  
Section 4.2

- 25 The following chart is *best* described as an example of which type of technical analysis chart?



- A A point and figure chart  
B A bar chart  
C A candlestick chart

C is correct. The chart is an example of a candlestick chart. A candlestick chart provides four prices per data point entry: the opening and closing prices and the high and low prices during the period (i.e., during a week). In a candlestick chart, a vertical line represents the range through which the security price traveled during the time period. The line is known as the wick or shadow. The body of the candle is shaded if the opening price was higher than the closing price, and the body is clear if the opening price was lower than the closing price.

A is incorrect; the chart is an example of a candlestick chart.

B is incorrect; the chart is an example of a candlestick chart.

---

Technical Analysis  
LOS b, d  
Section 3.1.3

- 26 The distribution of all the distinct possible values for a statistic when calculated from samples of the same size randomly drawn from the same population is *most* accurately referred to as:

- A a discrete uniform distribution.



- B a multivariate normal distribution.
- C the sampling distribution of a statistic.

C is correct. The sampling distribution of a statistic (like a sample mean) is defined as the probability distribution of a given sample statistic when samples of the same size are randomly drawn from the same population.

A is incorrect. Discrete uniform distribution refers to the distribution of equally likely outcomes.

B is incorrect. Multivariate normal distribution specifies the probabilities for a group of related random variables.

---

Sampling and Estimation  
LOS a  
Section 2.1

- 27 A US Treasury bill with a face value of \$100,000 and 140 days remaining to maturity is selling for \$98,000. The money market yield is *closest* to:
- A 5.25%.
  - B 5.14%.
  - C 5.32%.

A is correct. The money market yield,  $r_{MM} = \frac{360r_{BD}}{360 - (t)r_{BD}}$ , where

$r_{BD}$  = the annualized yield on a bank discount basis  
 $t = 140$  and is the number of days remaining to maturity

$$r_{BD} = (2,000/100,000)(360/140) = 0.0514 \text{ or } 5.14\%$$

$$r_{MM} = (360)(0.0514)/[(360 - (140)(0.0514)] = 0.05247 \sim \mathbf{5.25\%}$$

Alternatively,  $r_{MM} = \frac{F - S}{S} \times \frac{360}{t} = \frac{2,000}{98,000} \times \frac{360}{140} = \mathbf{5.25\%}$ , where S is the current selling price.

B is incorrect. This is  $r_{BD}$  as per calculation shown in A.

C is incorrect. If 365 is used instead of 360,

$$r_{MM} = \frac{F - S}{S} \times \frac{360}{t} = \frac{2,000}{98,000} \times \frac{365}{140} = 5.32\%$$

---

Discounted Cash Flow Applications  
LOS e  
Section 4

- 28 A bank offers an effective annual rate (EAR) of 12%. Assuming quarterly compounding, the stated annual interest rate is *closest* to:
- A 11.66%.
  - B 12.55%.
  - C 11.49%.

C is correct. First, find the periodic interest rate by rearranging the following formula:

$$\text{EAR} = (1 + \text{Periodic interest rate})^m - 1$$

$$12.00\% = (1 + \text{Periodic interest rate})^4 - 1$$

By solving for the periodic interest rate we obtain:

$$\text{Periodic interest rate} = \sqrt[4]{(12.00\% + 1)} - 1 = 2.873734\%$$

Then, find the stated annual interest rate as follows:

$$\begin{aligned} \text{Stated annual interest rate} &= \text{Periodic interest rate} \times m \\ &= 2.873734\% \times 4 \\ &= 11.49\% \end{aligned}$$

A is incorrect. It uses semiannual rather than quarterly compounding:

$$12.00\% = (1 + \text{Periodic interest rate})^2 - 1$$

By solving for the periodic interest rate we obtain:

$$\text{Periodic interest rate} = \sqrt[2]{(12.00\% + 1)} - 1 = 5.83\%$$

Then, find the stated annual interest rate as follows:

$$\text{Stated annual interest rate} = 5.83\% \times 2 = 11.66\%$$

B is incorrect. It uses the following formula:

$$\left(1 + \frac{12.00\%}{4}\right)^4 - 1 = 12.55\%$$

The Time Value of Money  
LOS c  
Section 3.2

- 29 An investor currently has a portfolio valued at \$700,000. The investor's objective is long-term growth, but she will need \$30,000 by the end of the year to pay her son's college tuition and another \$10,000 by year-end for her annual vacation. The investor is considering three alternative portfolios:

| Portfolio | Expected Return | Standard Deviation of Returns | Safety-First Ratio |
|-----------|-----------------|-------------------------------|--------------------|
| 1         |                 |                               | 0.2290             |
| 2         |                 |                               | 0.3300             |
| 3         | 14%             | 22%                           |                    |

Using Roy's safety-first criterion, which of the alternative portfolios *most likely* minimizes the probability that the investor's portfolio will have a value lower than \$700,000 at year-end?

- A Portfolio 1
- B Portfolio 3
- C Portfolio 2

B is correct. The investor requires a minimum return of  $(\$30,000 + \$10,000)/\$700,000$ , or 5.71%. Roy's safety-first model uses the excess portfolio's expected return over the minimum return and divides that excess by the standard deviation for that portfolio:

$$\text{Safety-first ratio} = [E(R_P) - R_L]/\sigma_P,$$

where

$E(R_P)$  = the expected return of portfolio  $P$

$R_L$  = the minimum return required by the investor

$\sigma_P$  = the standard deviation of returns of portfolio  $P$

| Portfolio | Safety-First Ratio              |
|-----------|---------------------------------|
| 1         | 0.2290                          |
| 2         | 0.3300                          |
| 3         | $(14\% - 5.71\%)/22\% = 0.3768$ |

The portfolio with the highest safety-first ratio minimizes the probability that the investor's portfolio will have a value lower than \$700,000 at year end. The highest safety-first ratio is associated with Portfolio 3: 0.3768.

A is incorrect. The investor requires a minimum return of  $(\$30,000 + \$10,000)/\$700,000$  or 5.71 percent. Roy's safety-first model uses the excess portfolio's expected return over the minimum return and divides that excess by the standard deviation for that portfolio:

$$\text{Safety-first ratio} = [E(R_P) - R_L]/\sigma_P,$$

where

$E(R_P)$  = is the expected return of portfolio  $P$

$R_L$  = the minimum return required by the investor

$\sigma_P$  = the standard deviation of returns of portfolio  $P$

| Portfolio | Safety-First Ratio              |
|-----------|---------------------------------|
| 1         | 0.2290                          |
| 2         | 0.3300                          |
| 3         | $(14\% - 5.71\%)/22\% = 0.3768$ |

The portfolio with the highest safety-first ratio minimizes the probability that the investor's portfolio will have a value lower than \$700,000 at year end. The highest safety-first ratio is associated with Portfolio 3: 0.3768.

C is incorrect. The investor requires a minimum return of  $(\$30,000 + \$10,000)/\$700,000$  or 5.71 percent. Roy's safety-first model uses the excess portfolio's expected return over the minimum return and divides that excess by the standard deviation for that portfolio:

$$\text{Safety-first ratio} = [E(R_P) - R_L]/\sigma_P,$$

where

$E(R_P)$  = is the expected return of portfolio  $P$

$R_L$  = the minimum return required by the investor

$\sigma_P$  = the standard deviation of returns of portfolio  $P$

| Portfolio | Safety-First Ratio              |
|-----------|---------------------------------|
| 1         | 0.2290                          |
| 2         | 0.3300                          |
| 3         | $(14\% - 5.71\%)/22\% = 0.3768$ |

The portfolio with the highest safety-first ratio minimizes the probability that the investor's portfolio will have a value lower than \$700,000 at year end. The highest safety-first ratio is associated with Portfolio 3: 0.3768.

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Common Probability Distributions  
LOS m  
Section 3.3

- 30** Which of the following is *most likely* to be an explanation of the power of a test? The power of a test is the probability of:
- A** rejecting the null when it is false.
  - B** not accepting the alternative when it is false.
  - C** a Type I error.

A is correct. The power of a test is the probability of correctly rejecting the null—that is, the probability of rejecting the null when it is false.

B is incorrect. The power of a test is the probability of correctly rejecting the null—that is, the probability of rejecting the null when it is false. If the alternative is not accepted when it is false, it means that a true null hypothesis is not rejected. Therefore, the probability of not accepting the alternative when it is false is not an explanation of the power of a test.

C is incorrect. The power of a test is the probability of correctly rejecting the null—that is, the probability of rejecting the null when it is false. The probability of a Type I error is equal to the probability of rejecting the null when it is true and is also known as the level of significance. Therefore, it is not an explanation of the power of a test.

---

Hypothesis Testing  
LOS c, d  
Section 2

- 31** A *z*-test regarding a mean is *most* appropriate when the variance is unknown and the:
- A** sample is not normally distributed.
  - B** central limit theorem is applicable.
  - C** sample is small and normally distributed.

B is correct. If the population variance is unknown and the sample is large, we can use a *z*-test relying on the central limit theorem.

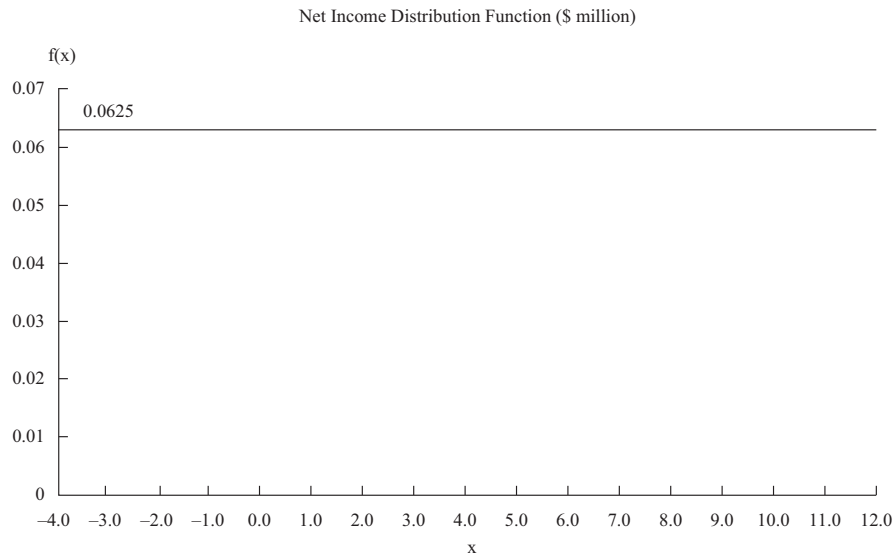
A is incorrect because a *t*-test is appropriate in the case of an unknown variance with moderate departures from normality.

C is incorrect because in the case of a hypothesis test of the population mean with an unknown variance, a test statistic is used when either 1) the sample is large, or 2) the sample is small but normally distributed, or approximately normally distributed. A *t*-test is typically used with small samples and an unknown population variance.

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Hypothesis Testing  
LOS g  
Section 3.1

- 32 A company forecasts that net income next year will range from a loss of \$4,000,000 to a gain of \$12,000,000. The graph shows the probability density function for this continuous distribution.



Which of the following statements is correct?

- A The probability of avoiding a loss is 0.25.
- B The expected net income is \$8,000,000.
- C The probability of incurring a loss of \$2,000,000 is 0.

C is correct. For a continuous uniform random variable, all outcomes are equally likely, and the probability of the random variable assuming any given fixed value, such as \$2,000,000, is zero.

A is incorrect because the probability of avoiding a loss is the area under the probability density function from zero to \$12,000,000 =  $P(0 \leq X \leq 12)$ . It includes an area with a base of 12 and a height of  $1/16$ . Base times height =  $12/16 = 75.0\%$  of the total rectangular distribution function.

B is incorrect because, as a continuous uniform random variable, the expected value (mean) for net income is the midpoint between the lowest and highest values:

$$\begin{aligned}
 \mu &= (a + b)/2 \\
 &= (-\$4,000,000 + \$12,000,000)/2 \\
 &= \$8,000,000/2 \\
 &= \$4,000,000
 \end{aligned}$$

Common Probability Distributions  
LOS h  
Section 3.1

- 33 Technical analysts following Elliott Wave Theory *most likely* use Fibonacci numbers to determine the expected:
- A size of future waves in a market trend.
  - B correlation between waves and technical indicators.
  - C number of waves in a grand supercycle.



A is correct. R.N. Elliott found that the ratio of the size of subsequent waves in a market trend was generally a ratio derived from Fibonacci numbers.

B is incorrect. Ratios based on Fibonacci numbers help predict the size of subsequent waves and are not related to determining correlation between waves and technical indicators.

C is incorrect. The number of waves in a cycle is always the same; it is the size of the wave that can be forecasted based on ratios established using Fibonacci numbers. The longest of the waves is called the grand supercycle.

Technical Analysis

LOS g

Section 4

**34** An expansionary fiscal policy is *most likely* associated with:

- A** crowding out of private investments.
- B** an increase in government spending on social insurance and benefits.
- C** an increase in capital gains tax rates.



A is correct. Expansionary policy increases government borrowing, which may divert private sector investment from taking place (resulting in an effect known as crowding out). A rise in capital gain tax rates is a form of contractionary fiscal policy. Rises in government spending on social insurance and benefits is a form of automatic stabilizer and not due to discretionary fiscal expansion.

C is incorrect. Rises in capital gains taxes is form of contractionary fiscal policy.

B is incorrect. Rises in government spending on social insurance and benefits is a form of automatic stabilizer and not due to discretionary fiscal expansion.

Monetary and Fiscal Policy

LOS p, q, s

Sections 3.1.1, 3.1.2, 3.1.3

**35** Holding the working-age population constant, if the labor force participation rate declines while the number of people employed remains unchanged, the unemployment rate will *most likely*:

- A** remain unchanged.
- B** increase.
- C** decrease.



C is correct. For a given working-age population, a decline in the labor force participation rate (often caused by an increase in discouraged workers) reduces the labor force. If the number of people employed remains the same while the labor force becomes smaller, the number of workers defined to be unemployed must be smaller and thus the unemployment rate lower.

The following example illustrates the direction of change:

|                                     | Initial Case   | After Change   |
|-------------------------------------|----------------|----------------|
| Working-age population              | 100            | 100            |
| Labor force = Employed + Unemployed | $60 + 20 = 80$ | $60 + 15 = 75$ |
| Labor force participation rate      | 80%            | 75%            |
| Unemployment rate                   | $20/80 = 25\%$ | $15/75 = 20\%$ |

Labor force participation rate = Labor force/Working age population  
Unemployment rate = Unemployed/Labor force

A is incorrect, as per table above.

B is incorrect, as per table above.

Understanding Business Cycles

LOS d

Section 4.1

36 An expansionary fiscal policy is *least likely* to include an increase in:

- A government expenditures.
- B budget deficit.
- C tax rates.

C is correct. An expansionary fiscal policy means that the government increases its purchases of goods and services and/or cuts tax rates to increase aggregate demand. Furthermore, an increase in the budget deficit would be associated with an expansionary fiscal policy.

B is incorrect. A rise in the budget deficit would be associated with an expansionary fiscal policy.

A is incorrect. An expansionary fiscal policy means that government increases its purchases of goods and services and/or cuts tax rates to increase aggregate demand.

Monetary and Fiscal Policy

LOS s

Sections 3.1.1–3.1.2

37 The total output in units and average selling prices in a hypothetical economy producing only two products, X and Y, is provided:

| Year | Product X      |                    | Product Y      |                    |
|------|----------------|--------------------|----------------|--------------------|
|      | Output (units) | Selling Price/unit | Output (units) | Selling Price/unit |
| 1    | 2,800          | €9                 | 2,000          | €47                |
| 2    | 3,000          | €11                | 1,800          | €52                |

If the implicit price deflator for GDP in Year 1 was 100, for Year 2, it is *closest* to:

- A 106.2.
- B 113.4.
- C 106.8.

B is correct.

| Year | Nominal GDP                                         | Real GDP                                                 |
|------|-----------------------------------------------------|----------------------------------------------------------|
| 1    | $(2,800 \times 9) + (2,000 \times 47) = 119,200$    | 119,200                                                  |
| 2    | $(3,000 \times 11) + (1,800 \times 52) = 126,600^*$ | $(3,000 \times 9) + (1,800 \times 47) = 111,600^\dagger$ |

\* Value of current output at current prices

† Value of current output at base year prices

$$\begin{aligned} \text{GDP Deflator} &= \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100 = \frac{126,600}{111,600} \times 100 = 113.4 \\ &= \frac{\text{Value of current output at current prices}}{\text{Value of current output at base year prices}} \times 100 \end{aligned}$$

A is incorrect. It compares growth in nominal GDP in the two years

$$126,600/119,200 \times 100 = 106.2$$

C is incorrect. It compares real GDP between years:

$$119,200/111,600 \times 100 = 106.8$$

Aggregate Output, Prices, and Economic Growth  
LOS a, c  
Section 2.1.2

- 38 A country with which of the following characteristics is *least likely* to face long-term GDP growth challenges?
- A A country with innovations in production processes
  - B A country with high labor quality
  - C A country with large natural resources

A is correct. The most important factor affecting economic growth is technology because it allows an economy to overcome the limits imposed by diminishing marginal returns. A country with innovations in the production process is least likely to face long-term GDP growth challenges compared with a country that relies on input growth, such as labor or natural resources.

C is incorrect. Natural resources are key inputs to GDP growth; however, technology is the most important factor that will help the country overcome the limits imposed by diminishing marginal returns. Even though natural resources are important, they are not necessary for a country to achieve a high level of income provided it can acquire the natural resources through trade.



B is incorrect. High labor quality is key input to GDP growth; however, technology is the most important factor that will help the country overcome the limits imposed by diminishing marginal returns.

Aggregate Output, Prices, and Economic Growth  
LOS o, n  
Section 4.2

39 A new technology that reduces employee illness will *most likely* decrease:

- A demand–pull inflation.
- B cost–push inflation.
- C cost–pull inflation.

B is correct. By reducing employee illness, the new technology will increase the output per hour per worker, which will decrease the unit labor cost. As the unit labor cost decreases, cost–push inflation decreases. The technology does not affect demand and accordingly should not affect demand–pull inflation.

A is incorrect because the technology decreases the cost–push inflation and does not affect the demand–pull inflation.

C is incorrect because cost–pull inflation is not a type of inflation.

Understanding Business Cycles  
LOS h  
Section 4.2.4

40 A research report produced by a dealer includes the following exchange rates:

|         | Spot Rate | Expected Spot Rate<br>in One Year |
|---------|-----------|-----------------------------------|
| USD/EUR | 1.3960    | 1.3860                            |
| USD/CAD | 1.0110    | 1.0300                            |
| EUR/GBP | 1.2850    | 1.2790                            |

The expected appreciation (%) of the Canadian dollar (CAD) relative to the British pound (GBP) is *closest* to:

- A –3.00
- B 3.09.
- C 0.70.

B is correct.

|         | Spot Rate | Expected Spot Rate in One Year | Appreciation: Expected/<br>Spot – 1 |
|---------|-----------|--------------------------------|-------------------------------------|
| USD/EUR | 1.3960    | 1.3860                         | –0.72%                              |
| USD/CAD | 1.0110    | 1.0300                         | 1.88%                               |
| EUR/GBP | 1.2850    | 1.2790                         | –0.44%                              |

(continued)

|          | Spot Rate | Expected Spot Rate in One Year                                                     | Appreciation: Expected/<br>Spot – 1 |
|----------|-----------|------------------------------------------------------------------------------------|-------------------------------------|
| CAD/GBP  | =         | $[(\text{USD}/\text{EUR}) \times (\text{EUR}/\text{GBP})]/(\text{USD}/\text{CAD})$ |                                     |
| CAD/GBP  | 1.7743    | 1.7211                                                                             | –2.99%                              |
| GBP/CAD* | 0.5636    | 0.5810                                                                             | 3.09%                               |

\* Canadian dollar is the base currency and the British pound is the price currency

$$\text{CAD/GBP} = [(\text{USD}/\text{EUR}) \times (\text{EUR}/\text{GBP})]/(\text{USD}/\text{CAD})$$

$$\text{Spot Rate of CAD/GBP} = (1.3960 \times 1.2850)/1.0110 = 1.7743$$

$$\text{Expected Spot Rate of CAD/GBP} = (1.3860 \times 1.2790)/1.0300 = 1.7211$$

To determine the appreciation of the Canadian dollar (CAD) relative to the British pound, the British pound (GBP) is the price currency and the Canadian dollar is the base currency, giving rise to the following calculation:

$$\frac{1/1.7211}{1/1.7743} - 1 = \frac{1.7743}{1.7211} - 1 = 3.09\%$$

B is incorrect. It calculates the spot and expected spot rates correctly, but uses GBP as the base currency: Appreciation of CAD/GBP =  $1.7211/1.7743 - 1 = -3.00\%$ .

C is incorrect. The appreciation of the individual exchange rates are summed:  $-0.72 + 1.88 - 0.44 = 0.72$ .

Currency Exchange Rates  
LOS d  
Section 3.2

- 41 A power generation company is a monopoly that has very high barriers to entry. The quantity demand ( $Q_D$ ) for its product is  $Q_D = 800 - 0.25 \times P$  (where  $P$  is price). The slope of the marginal revenue curve is *closest* to:
- A –0.25.
  - B –8.00.
  - C –4.00.

B is correct. Solve for  $P$  from the quantity demanded:

$$Q = 800 - 0.25P$$

$$P = 3,200 - 4Q$$

$$\text{TR} = P \times Q = 3,200Q - 4Q^2$$

$$\text{MR} = \Delta \text{TR} / \Delta Q = 3,200 - 8Q$$

Therefore, the slope of the curve is –8.

A is incorrect because it is the slope of the quantity demanded.

C is incorrect because it is the slope of the demand curve ( $-1/0.25$ ) when  $P$  is a function of  $Q$ .

The Firm and Market Structures  
LOS d  
Section 6.3

- 42 With respect to the relationship between output and costs in the short run, a decline in the marginal cost per unit *most likely* occurs at what level of production?
- A Low output
  - B Profit-maximizing output
  - C High output



A is correct. Marginal cost per unit, in the short run, decreases at low levels of output as a result of economies from greater specialization. At higher levels of output, however, it eventually increases because of the law of diminishing returns.

B is incorrect. Assuming a given price per unit (i.e., a flat marginal revenue curve), the profit-maximizing output must be above the minimum point on the marginal cost curve. That is, MC must be increasing at  $MR = MC$ .

C is incorrect. At higher levels of production, marginal cost eventually increases because of the law of diminishing returns.

---

Topics in Demand and Supply Analysis  
LOS d  
Section 3.2.3

- 43 Which of the following actions on the part of a central bank is *most* consistent with increasing the quantity of money?
- A Purchasing securities on the open market
  - B Increasing the central bank policy rate
  - C Increasing reserve requirements



A is correct. When a central bank purchases securities, bank reserves increase. The banks therefore have excess reserves and are able to increase their lending, increasing the money supply.

B is incorrect. The higher the policy rate, the higher the potential penalty that banks will have to pay to the central bank if they run short of liquidity, the greater will be their willingness to reduce lending, and the more likely the money supply will decrease.

C is incorrect. If the central bank increases required reserve ratios, the banks will decrease lending and the money supply will decline.

---

Monetary and Fiscal Policy  
LOS h, m  
Sections 2.3.2

- 44 The goods and services produced by UK citizens who are working in the United States are *most likely* included in the:
- A GDP of the United Kingdom and GNP of the United States.
  - B GNP of the United Kingdom and GDP of the United States.
  - C GNP of the United Kingdom and GNP of the United States.



B is correct. GDP measures the market value of all final goods and services produced by factors of production located within a country during a given period. GNP measures the market value of all final goods of services produced by factors of production supplied by residents of country, regardless of where they are located. Production by UK citizens working in the United States is included in the United Kingdom's GNP and the United States' GDP.

A is incorrect because the production of UK citizens in the United States should not be included in the GDP of the United Kingdom or in the GNP of the United States.

C is incorrect because the production of UK citizens should not be included in the GNP of the United States.

---

International Trade and Capital Flows

LOS a

Section 2.1

**45** A decrease in the price of a good will *most likely* be reflected in a:

- A** change in the intercept of the demand curve.
- B** change in the slope of the demand curve.
- C** downward movement along the demand curve.



C is correct. A decrease in the price of a good will cause consumers to demand more of a good and be reflected by a downward movement along the demand curve.

A is incorrect. A change in the intercept of the demand curve (a shift up or down without a change in slope) occurs when a variable other than own-price takes on a different value.

B is incorrect. The slope of the demand curve is the coefficient on  $Q$  (quantity) in the inverse demand curve, showing the change in  $P$  (price) for a 1-unit change in  $Q$ . A change in the slope of the demand curve occurs when the relationship between price and quantity changes.

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Topics in Demand and Supply Analysis

LOS a

Section 2.1

**46** Which of the following is *least likely* to be a general feature underlying the preparation of financial statements within the International Financial Reporting Standards (IFRS) Conceptual Framework?

- A** Matching
- B** Accrual basis
- C** Materiality



A is correct. The IFRS Conceptual Framework specifies a number of general features underlying the preparation of financial statements, including materiality and accrual basis. Matching is not one of those general features; it is a general principle of expense recognition.

B is incorrect. Accrual basis is one of the general features underlying the preparation of financial statements under the IFRS Framework

C is incorrect. Materiality is one of the general features underlying the preparation of financial statements under the IFRS Framework

Financial Reporting Standards  
LOS e  
Sections 5.5, 5.5.2  
Understanding Income Statements  
LOS d  
Section 4.1

- 47 Which of the following statements is *most* accurate with respect to the jurisdiction underlying financial reporting?
- A The requirement to prepare financial reports in accordance with specified accounting standards is the responsibility of standard-setting bodies.
  - B Regulatory authorities are typically private sector, self-regulated organizations.
  - C Standard-setting bodies have authority because they are recognized by regulatory authorities.

C is correct. Without the recognition of the standards by the regulatory authorities, such as the US SEC, private sector standard-setting bodies, such as the US Financial Accounting Standards Board, would have no authority.

A is incorrect. The requirement to prepare financial statements in accordance with specific accounting standards is the responsibility of regulatory authorities.

B is incorrect. Standard-setting bodies (not regulatory authorities) are typically private sector, self-regulated organizations.

Financial Reporting Standards  
LOS b  
Section 3

- 48 A company incurs the following costs related to its inventory during the year:

| Cost                                                          | ¥ millions |
|---------------------------------------------------------------|------------|
| Purchase price                                                | 100,000    |
| Trade discounts                                               | 5,000      |
| Import duties                                                 | 20,000     |
| Shipping of raw materials to manufacturing facility           | 10,000     |
| Manufacturing conversion costs                                | 50,000     |
| Abnormal costs as a result of waste material                  | 8,000      |
| Storage cost of finished goods prior to shipping to customers | 2,000      |

The amount charged to inventory cost (in millions) is *closest* to:

- A ¥177,000.
- B ¥185,000.
- C ¥175,000.

C is correct. The costs to include in inventories are all costs of purchase, costs of conversion, and other costs incurred in bringing the inventories to their present location and condition. It does not include abnormal waste costs or storage of finished product.

| Cost                                                | ¥ millions |
|-----------------------------------------------------|------------|
| Purchase price                                      | 100,000    |
| Minus trade discounts                               | -5,000     |
| Import duties                                       | 20,000     |
| Shipping of raw materials to manufacturing facility | 10,000     |
| Manufacturing conversion costs                      | 50,000     |
| Total inventory costs                               | 175,000    |

A is incorrect. It mistakenly includes the ¥2,000 storage cost:  $¥175,000 + ¥2,000 = ¥177,000$ .

B is incorrect. It mistakenly adds, not subtracts, the trade discounts of ¥5,000 from the purchase price:  $¥175,000 + ¥10,000 = ¥185,000$ .

Inventories  
LOS a  
Section 2

- 49 For which of the following assets is it *most* appropriate to test for impairment at least annually?
- A Land
  - B A patent with a legal life of 20 years
  - C A trademark with an indefinite expected life

C is correct. Intangible assets with indefinite lives need to be tested for impairment at least annually. Property, plant, and equipment (including land) and intangibles with finite lives are only tested if there has been a significant change or other indication of impairment.

A is incorrect. Property, plant, and equipment (including land) are not tested annually but only when significant events suggest a need to test for impairment.

B is incorrect. Intangible assets with finite lives are not tested annually but only when significant events suggest a need to test for impairment.

Understanding Balance Sheets  
LOS e  
Sections 4.1, 4.3  
Long-Lived Assets  
LOS i  
Sections 5.1, 5.2, 5.3

- 50 Interim reports *most likely*:
- A are audited.
  - B are issued semi-annually or quarterly.
  - C include a full set of financial statements and notes.

B is correct. Interim reports are provided semi-annually or quarterly, depending on applicable regulatory requirements.

A is incorrect. Interim reports are not audited.

C is incorrect. Interim reports generally present the four basic financial statements and condensed notes.

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Financial Statement Analysis: An Introduction

LOS e

Section 3.2

51 The following data are available on a company:

| Metric                                | Company |
|---------------------------------------|---------|
| Stock price per share                 | \$60.75 |
| Comprehensive income (millions)       | \$193.0 |
| Other comprehensive income (millions) | \$87.6  |
| Common shares outstanding (millions)  | \$46.5  |

On a net income basis, the company's P/E is *closest* to:

A 10.1.

B 14.6.

C 26.8.

C is correct.

$$\begin{aligned}
 \text{Net income} &= \text{Comprehensive income} - \text{Other comprehensive income} \\
 &= \$193.0 - \$87.6 \\
 &= \$105.4 \text{ million}
 \end{aligned}$$

$$\begin{aligned}
 \text{Net income per share (EPS)} &= \text{Net income} / \text{Common shares outstanding} \\
 &= \$105.4 / 46.5 \\
 &= \$2.27 \text{ million}
 \end{aligned}$$

$$\begin{aligned}
 \text{P/E} &= \text{Stock price} / \text{EPS} \\
 &= \$60.75 / \$2.27 \\
 &= 26.76
 \end{aligned}$$

A is incorrect. Other comprehensive income was incorrectly added to comprehensive income.

$$\begin{aligned}\text{Net Income} &= \text{Comprehensive Income} + \text{Other comprehensive income} \\ &= \$193.0 + \$87.6 \\ &= \$280.6\end{aligned}$$

$$\begin{aligned}\text{Net Income per share (EPS)} &= \text{Net income}/\text{Shares outstanding} \\ &= \$280.6/46.5 \\ &= \$6.03\end{aligned}$$

$$\begin{aligned}\text{P/E} &= \text{Stock Price}/\text{EPS} \\ &= \$60.75/\$6.03 \\ &= 10.07\end{aligned}$$

B is incorrect because the comprehensive P/E was used.

$$\begin{aligned}\text{Comprehensive income per share} &= \text{Comprehensive income}/\text{Shares outstanding} \\ &= \$193.0/46.5 \\ &= \$4.15\end{aligned}$$

$$\begin{aligned}\text{Comprehensive P/E} &= \text{Stock price}/\text{Comprehensive income per share} \\ &= \$60.75/\$4.15 \\ &= 14.64\end{aligned}$$

Understanding Income Statements  
LOS I  
Section 8  
Financial Analysis Techniques  
LOS b  
Section 5.1

- 52 Based on the presented information about a company's trade receivables, the bad debt expense (in £ millions) for 2014 is *closest* to:

| (£ millions)                    | 2014  | 2013  |
|---------------------------------|-------|-------|
| Accounts receivable, gross      | 6,620 | 4,840 |
| Allowance for doubtful accounts | 92    | 56    |
| Write-offs during the year      | 84    | 42    |

- A 84.  
B 36.  
C 120.

C is correct. The allowance for doubtful accounts increases by the bad debt expense recognized for the year and decreases by the amounts written off during the year.



|                                                   |              |
|---------------------------------------------------|--------------|
| Beginning balance allowance for doubtful accounts | £56 million  |
| Plus bad debt expense                             | ?            |
| Minus write-offs                                  | —£84 million |
| Ending balance allowance for doubtful accounts    | £92 million  |

Solve for bad debt expense = £120 million.

A is incorrect. It is just write offs, which is not the same as the expense.

B is incorrect. It is just the increase in the allowance for the year ( $92 - 56 = 36$ ) and ignores the write-offs that reduce the account.

Understanding Balance Sheets

LOS e

Section 3.1.3

53 The following data are available on a company:

| Current Year                          | (¥ millions) |
|---------------------------------------|--------------|
| Cash                                  | 114          |
| Inventory                             | 462          |
| Marketable securities (at fair value) | 23           |
| Property, plant, and equipment (net)  | 677          |
| Receivables                           | 231          |
| Current liabilities                   | 390          |
| <b>Liquidity Ratios in Prior Year</b> |              |
| Cash ratio                            | 0.37         |
| Current ratio                         | 2.19         |
| Quick ratio                           | 0.97         |

The value of the company's liquidity ratio that decreased the *most* in the current year, compared with the prior year, is the:

- A cash ratio.
- B quick ratio.
- C current ratio.

C is correct.

|              | Current Ratio                                                                          | Cash Ratio                         | Quick Ratio                                      |
|--------------|----------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------|
| Numerator    | Current assets<br>= Cash +<br>Marketable<br>securities +<br>Receivables +<br>Inventory | Cash +<br>Marketable<br>securities | Cash + Marketable<br>securities +<br>Receivables |
| Denominator  | Current liabilities                                                                    | Current<br>liabilities             | Current liabilities                              |
| Current year |                                                                                        |                                    |                                                  |

(continued)

|                     | Current Ratio                | Cash Ratio       | Quick Ratio            |
|---------------------|------------------------------|------------------|------------------------|
| Numerator           | $114 + 23 + 231 + 462 = 830$ | $114 + 23 = 137$ | $114 + 23 + 231 = 368$ |
| Denominator         | 390                          | 390              | 390                    |
| Ratio: Current year | 2.13                         | 0.35             | 0.94                   |
| Ratio: Prior year   | 2.19                         | 0.37             | 0.97                   |
| Change in ratio     | -0.06                        | -0.02            | -0.03                  |

A is incorrect. The cash ratio is mistakenly selected as the one with the greatest increase if calculated as cash/current liabilities (mistakenly excluding marketable securities from the numerator).

|                      | Current Year |
|----------------------|--------------|
| Cash                 | 114          |
| Current liabilities  | 390          |
| Incorrect Cash ratio | 0.29         |

The prior year cash ratio was 0.37, so the decrease in the cash ratio would be highest at -0.08.

B is incorrect. The quick ratio is mistakenly selected as the one with the greatest decrease if calculated as (cash + receivables)/current liabilities (mistakenly excluding marketable securities from the numerator).

|                       | Current Year      |
|-----------------------|-------------------|
| Cash + receivables    | $114 + 231 = 345$ |
| Current liabilities   | 390               |
| Incorrect quick ratio | 0.88              |

The prior year quick ratio was 0.97, so the decrease in the quick ratio would be highest at -0.09.

Understanding Balance Sheets  
LOS h  
Section 7.2  
Financial Analysis Techniques  
LOS b  
Section 4.3

- 54 The following information for a company was taken from its financial statements and the accompanying notes:

| Income Statement (\$ thousands) |        |       |
|---------------------------------|--------|-------|
| For Periods Ending 31 December  | 2014   | 2013  |
| Net sales                       | 11,159 | 8,895 |
| Cost of goods sold (COGS)       | 9,898  | 7,901 |

**Note 5. Inventories:** Inventories are reported on a last-in, first-out (LIFO) basis. The LIFO reserve was \$867 thousand and \$547 thousand at the end of 2014 and 2013, respectively. During 2014, the company liquidated certain LIFO inventories that had been carried at lower costs in prior years, and the effect of the liquidation was to decrease COGS by \$263 thousand. No LIFO liquidation occurred in 2013.

After adjusting for the LIFO liquidation in 2014, the change in gross profit margin compared with 2013 is *most likely*:

- A essentially unchanged.
- B higher by 2.5%.
- C lower by 2.3%.

C is correct.

$$\text{Gross profit margin} = (\text{Sales} - \text{COGS}) / \text{Sales} \times 100$$

$$\text{Gross profit under LIFO in 2014 (\$ thousands)} = 11,159 - 9,898 = 1,261$$

This figure arose in part from the LIFO liquidation, which decreased COGS by \$263,000 and hence increased gross profit.

Adjusting the gross profit downward by this amount gives adjusted gross profit (\$ thousands) of  $1,261 - 263 = 998$ .

$$\text{Adjusted gross profit margin in 2014} = (998 / 11,159) \times 100 = 8.9\%$$

$$\text{Gross profit margin in 2013} = (8,895 - 7,901) / 8,895 = 11.2\%$$

After adjusting for the LIFO liquidation, gross profit margin is lower by  $(11.2\% - 8.9\%) = 2.3\%$ .

A is incorrect. It makes no adjustment for the LIFO liquidation effect:  $1,261 / 11,159 = 11.3\%$ , which is  $11.3\% - 11.2\% = 0.1\%$  higher, unchanged as rounded.

B is incorrect. It adds the decrease in COGS:  $(1,261 + 263) / 11,159 = 13.7\%$ , which is  $13.7\% - 11.2\% = 2.5\%$  higher.

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Inventories

LOS e

Section 4.2

55 A company pays its workers on the 1st and the 15th of each month. Employee wages earned from the 15th to the 30th of September are *best* described as a(n):

- A accrued expense.
- B prepaid expense.
- C unearned expense.

A is correct. Wage expenses that have been incurred but not yet paid are an example of an accrued expense: a liability that has not yet resulted in a cash payment.

B is incorrect. Prepaid expenses involve a cash payment made before recognizing an expense.

C is incorrect. Erroneous term.

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Financial Reporting Mechanics

LOS e

Section 5.1

56 A portion of a company's balance sheet appears in the following table (euros in millions):

|                       |         |
|-----------------------|---------|
| Cash                  | 4,000   |
| Marketable securities | 17,000  |
| Accounts receivable   | 225,000 |
| Inventory             | 229,000 |
| Total current assets  | 475,000 |
| Current liabilities   | 339,000 |

The company's quick ratio is *closest* to:

- A 1.40.
- B 0.06.
- C 0.73.

C is correct. The quick ratio is

$$\frac{\text{Cash} + \text{Marketable securities} + \text{Accounts receivable}}{\text{Current liabilities}} = \frac{4,000 + 17,000 + 225,000}{339,000} = 0.73$$

A is incorrect. It is the current ratio: (Current assets/Current liabilities) = (475,000/339,000) = 1.40.

B is incorrect. It is the cash ratio: (Cash + Marketable Securities)/Current liabilities = (4,000 + 17,000)/339,000 = 0.06.

Understanding Balance Sheets  
LOS h  
Section 7.2

57 The following financial statement data are available for a company:

| Metric                     | £ thousands |
|----------------------------|-------------|
| Net income                 | 500         |
| Depreciation               | 150         |
| Cash flow from operations  | 600         |
| Free cash flow to the firm | 300         |
| Beginning total assets     | 4,000       |
| Ending total assets        | 6,000       |
| Ending cash balance        | 50          |
| Book value                 | 3,000       |

The company's cash return on assets ratio is *closest* to:

- A 12%.
- B 10%.
- C 13%.

A is correct.

$$\frac{\text{Cash return on assets} = (\text{Cash flow from operations} / \text{Average total assets})}{600} = 12\%$$

$$\frac{1}{2} \times (6,000 + 4,000)$$

B is incorrect. It uses cash flow from operations (CFO)/ending total assets in the denominator, OR free cash flow (FCF)/book value.

Incorrect cash return on assets = 10%

CFO/Ending total assets = 600/6,000

FCF/Book value = 300/3,000

C is incorrect. It incorrectly uses net income plus depreciation in the numerator OR uses cash flow from operations (CFO) + ending cash balance in the numerator.

Incorrect cash return on assets = 13%

(Net income + Depreciation)/Average total assets = 650/5,000

(CFO + Ending cash balance)/Average total assets = 650/5,000

Understanding Cash Flow Statements  
LOS i  
Section 4.4

**58** The *most* appropriate statement about financial ratio analysis is that it has limited use as an analytical tool for:

- A** providing insights into microeconomic relationships within a company that help analysts project earnings.
- B** evaluating management.
- C** comparing companies that use different accounting methods.

C is correct. Financial ratio analysis is limited by the use of alternative accounting methods. Accounting methods play an important role in the interpretation of financial ratios. The lack of consistency across companies makes comparability difficult to analyze and limits the usefulness of ratio analysis.

A is incorrect. Financial ratios provide insights into microeconomic relationships within a company that help analysts project earnings and free cash flow.

B is incorrect. Financial ratios are useful in providing evaluation of management.

Financial Analysis Techniques  
LOS a  
Section 3.1.2

**59** Selected information for a company is provided.

|                    | \$ millions |
|--------------------|-------------|
| Sales              | 4,800       |
| Cost of goods sold | 2,880       |

(continued)

|                     | \$ millions |
|---------------------|-------------|
| Purchases           | 2,940       |
| Average receivables | 625         |
| Average inventory   | 710         |
| Average payables    | 145         |

The company's cash conversion cycle (in days) is *closest* to:

- A 84.
- B 138.
- C 120.

C is correct.

Cash conversion cycle = Days sales outstanding + Days of inventory on hand  
– Days of payables

|          | Accounts Receivable: Days<br>Sales Outstanding (DSO)  | Inventory: Days on Hand<br>(DOH)                         | Accounts Payables: Days in<br>Payables         |
|----------|-------------------------------------------------------|----------------------------------------------------------|------------------------------------------------|
| Turnover | (Sales/Average receivables)<br>4,800/625 = 7.68 times | (Cost of goods sold/Inventory)<br>2,880/710 = 4.06 times | (Purchases/Payables)<br>2,940/145 = 20.3 times |
| In days  | 365/7.68 = 48 days                                    | 365/4.06 = 90 days                                       | 365/20.3 = 18 days                             |

$$\begin{aligned}\text{Cash conversion cycle} &= \text{DSO} + \text{DOH} - \text{Days in payables} \\ &= 48 + 90 - 18 \\ &= 120 \text{ days}\end{aligned}$$

A is incorrect. It calculates DOH based on sales, not CGS:  $4,800/710 = 6.76$  times;  $365/6.76 = 54$ ;  $48 + 54 - 18 = 84$ .

B is incorrect. It does not subtract the days in payables:  $48 + 90 = 138$  days.

Financial Analysis Techniques  
LOS b  
Sections 4.2, 4.3.1  
Working Capital Management  
LOS c  
Section 2.2

- 60 An analyst is completing her analysis of three companies in the same industry. Which of these companies is the *most* solvent?

| £ (in thousands)        | Company A | Company B | Company C |
|-------------------------|-----------|-----------|-----------|
| Total debt              | £10,087   |           |           |
| Total assets            | £28,087   |           |           |
| EBIT                    | £3,903    |           |           |
| Interest payments       | £847      |           |           |
| Debt-to-assets ratio    |           | 37%       | 34.2%     |
| Interest coverage ratio |           | 5.34      | 5.55      |

- A Company C

- B Company B  
C Company A

A is correct.

| Company A               | Calculation   | Ratio |
|-------------------------|---------------|-------|
| Debt-to-Assets Ratio    | 10,087/28,087 | 35.9  |
| Interest Coverage Ratio | 3,903/847     | 4.61  |

| Ratio             | Company A | Company B | Company C | Comments                                      |
|-------------------|-----------|-----------|-----------|-----------------------------------------------|
| Debt-to-Assets    | 35.9%     | 37%       | 34.2%     | Lower is more solvent; C is the most solvent. |
| Interest Coverage | 4.61      | 5.34      | 5.55      | Higher is better; C is the most solvent.      |

*Conclusion:* Company C is the most solvent based on both ratios.

B is incorrect. Company B has a debt-to-assets ratio higher than that of Company A.

C is incorrect. Company C is more solvent than Company A considering both ratios.

Non-Current (Long-Term) Liabilities  
LOS k  
Section 5

- 61 Which of the following events is *most likely* to lead management to make biased accounting choices?
- A Changing the company's fiscal year  
B Changing the company's audit firm  
C Changing the company's CEO

C is correct. A company might report lower earnings in the CEO's first year to create a positive trajectory for future periods. Alternatively, there could be motivation to report higher earnings under the new CEO. Either situation could lead to biased accounting choices.

A is incorrect. Changing the fiscal year is not likely to lead to biased reporting.

B is incorrect. Changing the audit firm is not likely to lead to biased reporting.

Financial Reporting Quality  
LOS d  
Sections 2.3 and 3.1

- 62 A company using IFRS reports its interest payments on long-term debt as a financing activity. If the company reported under US GAAP, the *most likely* effect would be a:
- A higher cash flow from operations.  
B higher cash flow from financing activities.  
C lower cash flow from investing activities.



B is correct. Interest payments can be reported either as operating or financing cash flow under IFRS, but they can be reported only as operating cash flow under US GAAP. The interest payment was originally reported as financing activity under IFRS, but under US GAAP it would be an operating activity. Therefore, under US GAAP, cash flow from financing activities would be higher and operating cash flows lower by the same amount.

A is incorrect. Under US GAAP, the interest payment would be a cash outflow from operating activities and thus decrease the operating cash flow.

C is incorrect. Under US GAAP, the interest payment would be a cash outflow from operating activities, and cash flow from investing activities would stay the same.

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Understanding Cash Flow Statements

LOS c

Sections 2.1, 2.2

- 63** Two pharmaceutical companies, Company A and Company B, internally develop drugs and drug analytics software. Company A reports in accordance with IFRS, and Company B reports in accordance with US GAAP. Which of the following statements is *most* accurate regarding the development costs of the drug patents and software development?
- A** Company A can capitalize the development costs related to software development if it meets certain criteria.
  - B** Company B can capitalize the development costs related to drug development if it meets certain criteria.
  - C** Both companies must expense all development costs related to these intangible assets.



A is correct. IFRS allows development costs to be capitalized if certain criteria are met. Unlike US GAAP, under IFRS capitalization is not restricted to software development.

B is incorrect. Under US GAAP, capitalization of development costs is restricted to software development only. Development costs related to tangible assets other than software must be expensed regardless.

C is incorrect. Both IFRS and US GAAP allow certain development costs to be capitalized if certain criteria are met.

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Long-Lived Assets

LOS a, b

Sections 2.2, 2.5

- 64** A company that prepares its financial statements in accordance with IFRS incurred and capitalized €2 million of development costs during the year. These costs were fully deductible immediately for tax purposes, but the company is depreciating them over two years for financial reporting purposes. The company has a long history of profitability, which is expected to continue. Which is the *most* appropriate way for an analyst to incorporate the differential tax treatment in his analysis? He should include it in:
- A** equity when calculating the company's return-on-equity ratio.
  - B** liabilities when calculating the company's debt-to-equity ratio.
  - C** liabilities when calculating the company's current ratio.



B is correct. The different treatment for tax purposes and financial reporting purposes is a temporary difference and would create a deferred tax liability. Deferred tax liabilities should be classified as debt if they are expected to reverse with subsequent tax payments. The long history of profitability implies the company will likely be paying taxes in the following years, and hence an analyst could reasonably expect the temporary difference to reverse. Under IFRS, all deferred tax liabilities are non-current.

A is incorrect. It would be included in equity only if there were no expectation of subsequent tax payment, which is not likely given the history of profitability.

C is incorrect. Under IFRS, all deferred tax liabilities are non-current, so it would not be included in the current ratio even though it is expected to reverse next year.

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Income Taxes  
LOS b, i  
Sections 2.2, 7

- 65 A company owns an office building that it purchased in 2011 for \$1,000,000. The real estate market has been volatile in the last few years. The company uses the revaluation model as allowed by IFRS, and the following table shows the building's fair market values since 2011:

| Year | Year-End Fair Market Value (\$ thousands) |
|------|-------------------------------------------|
| 2011 | 1,000                                     |
| 2012 | 600                                       |
| 2013 | 800                                       |
| 2014 | 1,300                                     |

The net impact (in thousands) on the 2014 net income will *most likely* be an increase of:

- A \$200.
- B \$500.
- C \$300.

A is correct. The revaluation model per IFRS allows the asset to be carried at fair value. If the revaluation decreases the value, as it does here from 2011 to 2012, then later increases in value to the extent that they reverse the losses are recognized in net income: from \$800 to \$1,000 = \$200.

Any increase in excess of the reversal will be recorded directly in equity in a revaluation surplus account and not on the income statement: here,  $1,300 - 1,000 = 300$ .

B is incorrect. Not the entire gain ( $1,300 - 800$ ) should be booked to the income statement. Only the portion previously booked to the income statement should be booked to the income statement.

C is incorrect. This is the amount above the original purchase price, and it is the portion that would go to revaluation surplus, not NI:  $300,000 = 1,300,000 - 1,000,000$ .

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Long-Lived Assets  
LOS h  
Section 4

- 66 Under the indirect method of presenting operating cash flows, which action to alter the cash flow from operations will be most difficult to detect?
- A Defer payment of a current liability
  - B Transact with an unconsolidated special purpose entity
  - C Change inventory costing from FIFO to weighted average

B is correct. Unconsolidated special purpose entities are outside of the view of investors. Transacting with such an entity may initially produce the appearance of a positive or negative cash flow for the controlling company. Ultimately, this transaction will most likely be reversed along with the appearance of the initial cash flow.

A is incorrect because an examination of cash provided by operating activities will reveal that the increase in cash due to the deferred payment is offset by a comparable increase in accounts payable.

C is incorrect because changes in inventory accounting may affect gross profit and therefore net income, but with an opposite effect on the ending inventory value. Together these effects would likely have an offsetting impact on the appearance of cash generated by operating activities. One possible exception might be the effect on derived expenses such as the provision for income tax.

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Financial Reporting Quality  
LOS h  
Section 4.2.1 and 4.2.2

- 67 A credit analyst is *most likely* to place more focus on:
- A operating leverage than financial leverage.
  - B cash flows than accrual income.
  - C upside potential than downside risk.

B is correct. Credit analysts are particularly focused on assessing debt-paying ability, which is generated from cash flows and not from accrual-based measures, such as net income. The return to debtholders is limited by contract, so upside potential is less important than concern for loss.

Financial leverage is a major concern for credit analysts because it is associated with a greater chance of default. Operating leverage, which measures the volatility of operating income as a result of fixed costs, is a lesser concern.

A is incorrect. Credit analysts are more concerned with financial leverage because higher financial leverage is associated with higher levels of default.

C is incorrect. Debt providers are more concerned with downside risk because they are entitled only to contractual cash flows. Additional upside potential does not result in additional return.

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Financial Statement Analysis: Applications  
LOS c  
Section 4

- 68 Denson Corporation issued \$5,000,000 of five-year bonds at a discount. After three years, the company calls the bonds at 101 when the bond's carrying value is \$4,950,000. The company will realize a:

- A loss of \$50,000.
- B gain of \$50,000.
- C loss of \$100,000.

C is correct. The call price is \$5,050,000 ( $\$5,000,000 \times 1.01$ ) for the issue, and the carrying value is \$4,950,000, resulting in a \$100,000 loss ( $\$4,950,000 - \$5,050,000$ ).

A is incorrect because the carrying value of the issue should be used to calculate the loss instead of the par value:  $\$5,000,000 - \$5,050,000$ .

B is incorrect because it is calculated as the difference between the call price and the par value:  $\$5,050,000 - \$5,000,000$ .

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Non-Current Long-Term Liabilities

LOS c

Section 2.4

- 69 Assume Company X recognizes contract revenues using the percentage-of-completion method and that it enters into the following five-year contract:

|                         |              |
|-------------------------|--------------|
| Total Five Year Revenue | \$10,000,000 |
| Annual Costs            |              |
| Year 1                  | \$500,000    |
| Year 2                  | \$800,000    |
| Year 3                  | \$900,000    |
| Year 4                  | \$1,500,000  |
| Year 5                  | \$1,300,000  |
| Total                   | \$5,000,000  |

Company X's reported revenue in Year 5 will be:

- A \$10,000,000.
- B \$2,600,000.
- C \$6,300,000.

B is correct. Company X's reported revenue in Year 5 will be \$2,600,000, calculated as:  
Percentage-of Completion-Method, Year 5:

$$\begin{aligned}
 \text{Cost in a year/Total cost} &= \text{Annual percent of revenue to recognize} \\
 &= \$1,300,000/\$5,000,000 \\
 &= 26\% \\
 \text{Revenue} &= 26\% \times \$10,000,000 \\
 &= \$2,600,000
 \end{aligned}$$

A is incorrect because \$10,000,000 is the reported revenue recognized using the completed contract method under US GAAP (when the contract cannot be measured reliably and it is probable that costs will be recovered). Under these principles, no revenue may be recognized until the contract is complete. Thus, all revenue is recognized in Year 5.

C is incorrect because \$6,300,000 is the reported revenue recognized using the completed contract method under IFRS (when the contract cannot be measured reliably and it is probable that costs will be recovered). Under these standards, revenue may be recognized to the extent of contract costs incurred, as:

|                           |              |
|---------------------------|--------------|
| Total Five Year Revenue   | \$10,000,000 |
| Annual Costs              |              |
| Year 1                    | \$500,000    |
| Year 2                    | \$800,000    |
| Year 3                    | \$900,000    |
| Year 4                    | \$1,500,000  |
| Year 5                    | \$1,300,000  |
| Total                     | \$5,000,000  |
| Annual Revenue Recognized |              |
| Year 1                    | \$500,000    |
| Year 2                    | \$800,000    |
| Year 3                    | \$900,000    |
| Year 4                    | \$1,500,000  |
| Year 5                    | \$6,300,000  |
|                           | \$10,000,000 |

Understanding Income Statements  
LOS c  
Section 3.2.1

70 Given the following information about a firm:

- debt-to-equity ratio (D/E) of 50%
- tax rate of 40%
- cost of debt of 8%
- cost of equity of 13%

the firm's weighted average cost of capital (WACC) is *closest to*:

- A 8.9%.
- B 7.5%.
- C 10.3%.

C is correct. Convert D/E to the weight for debt:

$$\frac{1}{3} = \frac{D/E}{(1 + D/E)} = \frac{0.50}{(1 + 0.50)}$$

The weight for equity is one minus the weight of debt:  $2/3 = 1 - (1/3)$ .

$$\begin{aligned} \text{WACC} &= \text{weight of debt} \times \text{cost of debt} \times (1 - \text{tax rate}) + \text{weight of equity} \\ &\quad \times \text{cost of equity} \\ &= (1/3) \times 0.08 \times (1 - 0.40) + (2/3) \times 0.13 \\ &= 0.1026 = 10.3\% \end{aligned}$$

A is incorrect because it uses a 50% weight for debt and equity.

B is incorrect because it uses (2/3) for the weight of debt and (1/3) for the weight of equity.

---

Cost of Capital  
LOS a, b  
Sections 2, 2.1, 2.2

- 71** A company has 100 million shares outstanding. The share price of a company's stock is £15 just prior to announcing a £100 million expansionary investment in a new plant, and the company estimates that the present value of future after-tax cash flows will be £150 million. Analysts, however, estimate that the new plant's profitability will be lower than the company's expectations. The company's stock price will *most likely*:
- A** drop below £15 per share due to the cannibalization of revenue from the new plant.
  - B** increase by less than £0.50 per share.
  - C** increase by the new plant's net present value per share.



B is correct. The value of a company is the value of its existing investments plus the net present values of all of its future investments. The NPV of this new plant is £150 million – £100 million = £50 million. The price per share should increase by NPV per share or £50 million/100 million shares = £0.50 per share. As the new plant's profitability is less than expectations, the NPV per share (and hence the increase in the stock price) should therefore be slightly below £0.50 per share.

A is incorrect. It is only new plant's profitability that is below the average not the overall. The company value should not fall below £15 per share, all things being equal.

C is incorrect. See the above calculation.

---

Capital Budgeting  
LOS f  
Section 4.10

- 72** A company's optimal capital budget *most likely* occurs at the intersection of the:
- A** net present value and internal rate of return profiles.
  - B** marginal cost of capital and investment opportunity schedule.
  - C** marginal cost of capital and net present value profiles.



B is correct. The point at which the marginal cost of capital intersects the investment opportunity schedule is the optimal capital budget.

A is incorrect. The point at which the marginal cost of capital intersects the investment opportunity schedule is the optimal capital budget.

C is incorrect. The point at which the marginal cost of capital intersects the investment opportunity schedule is the optimal capital budget.

---

Capital Budgeting  
LOS e  
Section 4.7  
Cost of Capital

(continued)

LOS d  
Section 2.3

73 The following information is available for a firm:

|                        |             |
|------------------------|-------------|
| Sales price per unit   | €85         |
| Variable cost per unit | €65         |
| Fixed operating costs  | €50 million |
| Fixed financial costs  | €30 million |

The firm's breakeven quantity of sales (in million units) is *closest* to:

- A 4.0.
- B 2.5.
- C 1.0.

A is correct. Breakeven quantity of sales,

$$Q_{BE} = \frac{F + C}{P - V}$$

$$= (\text{€50 million} + \text{€30 million}) / (\text{€85} - \text{€65}) = 4.0 \text{ million units}$$

B is incorrect. If operating breakeven units is calculated, the answer will be  $Q_{OBE} = F / (P - V) = (\text{€50 million}) / (\text{€85} - \text{€65}) = 2.5 \text{ million}$ .

C is incorrect. If there is a wrong signage in the numerator, the answer will be  $= (\text{€50 million} - \text{€30 million}) / (\text{€85} - \text{€65}) = 1.0 \text{ million units}$ .

Measures of Leverage  
LOS d  
Section 3.6

74 Which of the following situations is the *least likely* reason why the marginal cost of capital schedule for a company rises as additional funds are raised?

- A The company seeks to issue less senior debt because it violates the debt incurrence test of an existing debt covenant.
- B The cost of additional funds from various sources rises as higher levels of financing are achieved.
- C The company deviates from its target capital structure because of the economies of scale associated with flotation costs and market conditions.

B is correct. The WACC does not necessarily increase as more funds are being raised. Higher amounts of funding would not change the WACC if everything were in proportion to the old target capital structure. It is the changes in relative proportions of sources of funding that could make a difference because of interest deductibility and financial risk.

A is incorrect. The debt incurrence test may restrict a company's ability to incur additional debt at the same seniority based on financial tests or conditions. They will have to issue a less senior debt (or even equity) which would have higher cost.

C is incorrect. A company does not necessarily raise more funds according to its target capital structure because of the economies of scale in raising new capital and market conditions. These short-run deviations are due to the “lumpiness” of security issuance. The marginal cost of capital may increase, reflecting these deviations.

---

Cost of Capital  
LOS k  
Section 4.3

- 75** A class of noncallable, nonconvertible preferred stock was issued at \$45.00 per share with a dividend of \$5.25. The preferred stock is now trading at \$60.00 per share. Earnings of the company are growing at 3.00%. The cost of preferred stock is *closest* to:
- A** 11.7%.
  - B** 8.8%.
  - C** 5.8%.

B is correct. The cost of preferred stock is:

$$r_p = \frac{D_p}{P_p}$$

where:

$r_p$  = the cost of preferred stock  
 $D_p$  = the preferred stock dividend per share  
 $P_p$  = the current preferred stock price per share

The cost of preferred stock is  $5.25/60.00 = 8.75\%$

C is incorrect. If the dividend growth rate of 3% is subtracted from 8.75% (as per the Gordon growth model), the conclusion will be 5.75%.

A is incorrect. If the issuance cost is being used instead of current price, it will lead to a wrong conclusion of 11.67%.

---

Cost of Capital  
LOS g  
Section 3

- 76** In order to maintain an adequate net daily cash position, a company is *least likely* to:
- A** monitor access to borrowing facilities.
  - B** forecast depreciation and accruals.
  - C** predict the business cycles and seasonal effects.

B is correct. Accruals are paid at a later date, and depreciation is a noncash expense.

A is incorrect. Most companies maintain a cash buffer as a protection from unexpected cash need or to provide the financial flexibility. The size of the buffer depends on several influences including the company's ability to access other liquidity sources or borrowing facilities.

C is incorrect. Correctly predicting the peak need caused by seasonality or other non-operating activities such as M&A or major capital expenditures can help the company to forecast cash required to cover the need.

Working Capital Management  
LOS d  
Sections 3.1–3.2

77 Assume a 365-day year and the following information for a company:

|                    | Current Year | Previous Year |
|--------------------|--------------|---------------|
| Sales              | \$12,000     | \$10,000      |
| Cost of goods sold | \$9,000      | \$7,500       |
| Inventory          | \$1,200      | \$1,000       |
| Accounts payable   | \$600        | \$600         |

The firm's days of payables for the current year is *closest* to:

- A 23.8.
- B 18.3.
- C 24.9.

A is correct. The number of days of payables is

$$\begin{aligned}\frac{\text{Accounts payable}}{(\text{Purchases}/365)} &= \frac{\text{Accounts payable}}{[(\Delta \text{ inventory} + \text{COGS})/365]} \\ &= \frac{\$600}{[(\$1,200 - \$1,000 + \$9,000)/365]} \\ &= 23.8\end{aligned}$$

B is incorrect because it uses sales instead of cost of goods sold and ignores the change in inventory:  $600/(12,000/365) = 18.3$ .

C is incorrect because it miscalculates purchases:  $600/[(1,000 - 1,200 + 9,000)/365] = 24.9$ .

Financial Analysis Techniques  
LOS b  
Section 4.2.2  
Working Capital Management  
LOS c, f  
Section 7.3

78 A company needs to nominate an independent director for its board of directors. The following candidates are being considered:

- A retired senior management official of the company
- A representative of a pension fund that owns 10% of the company's shares
- A former government employee who was involved in regulating the industry

Based on good corporate governance practices, which candidate is the *most* appropriate nominee?

- A The pension fund representative



- B The retired senior management official
- C The former government employee

C is correct. Independent directors must not have material relationships with the company with regard to employment, ownership, or remuneration. The former government employee best meets these criteria.

B is incorrect. The retired senior management officer has been employed by the company and likely still receives remuneration from the company in the form of a pension. He may also be reluctant to be critical of his own prior decisions or those of his co-managers.

A is incorrect. The pension fund has a material ownership relationship with the company, so its representative cannot be considered independent.

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Corporate Governance and ESG: An Introduction  
LOS f  
Section 5.1

- 79 Consider a portfolio with two assets. Asset A comprises 25% of the portfolio and has a standard deviation of 17.9%. Asset B comprises 75% of the portfolio and has a standard deviation of 6.2%. If the correlation of these two investments is 0.5, the portfolio standard deviation is *closest* to:
- A 6.45%.
  - B 7.90%.
  - C 9.13%.

B is correct. The standard deviation of a two-asset portfolio is given by the square root of the portfolio's variance:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$$

Using this formula, the existing standard deviation is calculated as follows:

$$\sqrt{0.25^2 \times 0.179^2 + 0.75^2 \times 0.062^2 + 2 \times 0.75 \times 0.25 \times 0.5 \times 0.179 \times 0.062} = 7.90\%$$

A is incorrect because it leaves off the correlation term in the equation:

$$\sqrt{0.25^2 \times 0.179^2 + 0.75^2 \times 0.062^2} = 6.45\%$$

C is incorrect because it is the weighted average standard deviation of the two assets given:  $(0.25 \times 0.179 + 0.75 \times 0.062) = 9.125\%$ .

---

Portfolio Risk and Return: Part I  
LOS e, f  
Section 2.3.3

- 80 The risk-free rate is 5%, and the market risk premium is 8%. If the beta of TRL Corp. is 1.5, based on the capital asset pricing model (CAPM), the expected return of TRL's stock is *closest* to:
- A 17.0%.
  - B 9.5%.
  - C 15.5%.

A is correct. Using the CAPM relationship of  $E(R_i) = R_f + [E(R_m) - R_f]\beta_i$ , we can estimate the expected return as:  $E(R_i) = 0.05 + (0.08)(1.5) = 17.0\%$ .

B is incorrect because the expected return is computed as  $0.05 + (0.08 - 0.05)(1.5) = 9.5\%$ .

C is incorrect because the expected return is computed as  $0.08 + (0.05)(1.5) = 15.5\%$ .

---

Portfolio Risk and Return: Part II  
LOS g  
Section 4.2

- 81 Following its decision to divest its non-core assets, analysts expect HCL Corp's standard deviation of returns to rise to 30% and its correlation with the market portfolio to remain unchanged at 0.8. The risk-free rate and the market risk premium are expected to remain unchanged at 6% and 8%, respectively. The market portfolio's standard deviation of returns, however, is expected to decrease to 15%. The firm's expected return after the restructure is *closest* to:
- A 9.2%.
  - B 17.6%.
  - C 18.8%.

C is correct. We first compute the firm's beta using

$$\beta_i = \frac{\rho_{i,m}\sigma_i}{\sigma_m}$$

The beta is

$$\beta_i = \frac{0.8(0.30)}{0.15} = 1.6$$

The expected return is computed using

$$E(R_i) = R_f + [E(R_m) - R_f]\beta_i$$

$$\text{So, } E(R_i) = 0.06 + (0.08)1.6 = 18.8\%.$$

A is incorrect because the beta is incorrectly calculated as  $[(0.8 \times 0.15)/0.30] = 0.4$ , resulting in an expected return of  $0.06 + (0.08)0.4 = 9.2\%$ .

B is incorrect because the expected return is incorrectly calculated as  $0.08 + (0.06)1.6 = 17.6\%$ .

---

Portfolio Risk and Return: Part II  
LOS e, g  
Sections 3.2 and 4.2

- 82 Which of the following types of institutions is *most likely* to have a long investment time horizon and a higher level of risk tolerance?
- A An endowment
  - B An insurance company
  - C A bank

A is correct. Endowments have a long investment time horizon and a high level of risk tolerance.

C is incorrect. Banks have a short investment time horizon and a low level of risk tolerance.

B is incorrect. Insurance companies have a short investment time horizon and a low level of risk tolerance.

---

Portfolio Management: An Overview

LOS b

Section 3.2

**83** Which of the following factors is *least likely* to affect an individual's ability to take risk?

**A** Expected income

**B** Personality type

**C** Time horizon

B is correct. An individual's ability to take risk is affected by such factors as time horizon and expected income. Personality type is most likely to affect an individual's willingness to take risk.

A is incorrect. Expected income will likely impact an individual's ability to take risk.

C is incorrect. Time horizon will likely impact an individual's ability to take risk.

---

Basics of Portfolio Management and Construction

LOS d

Section 2.2.1

**84** The covariance of the assets in the following portfolio is *closest* to:

|                  | Asset 1 | Asset 2 | Asset 1 vs. Asset 2 |
|------------------|---------|---------|---------------------|
| Correlation      |         |         | 0.8                 |
| Portfolio weight | 0.6     | 0.4     |                     |
| Variance         | 3.5%    | 1.5%    |                     |

**A** 1.8%

**B** 0.4%

**C** 2.3%

A is correct. The covariance is calculated from the standard deviations of the two assets and their correlation. The portfolio weights are not relevant.

$$\begin{aligned}\text{Cov} &= \rho_{1,2} \times \sigma_1 \times \sigma_2 \\ &= \rho_{1,2} \times \sqrt{\text{var}_1} \times \sqrt{\text{var}_2} \\ &= 0.8 \times \sqrt{3.5\%} \times \sqrt{1.5\%} \\ &= 1.8\%\end{aligned}$$

B is incorrect. The covariance is calculated from the standard deviations of the two assets and their correlation. The portfolio weights are not relevant.

$$\begin{aligned}&= \rho_{1,2} \times w_1 \times \sigma_1 \times w_2 \times \sigma_2 \\ &= \rho_{1,2} \times w_1 \times \sqrt{\text{var}_1} \times w_2 \times \sqrt{\text{var}_2} \\ &= 0.8 \times 0.6 \times \sqrt{3.5\%} \times 0.4 \times \sqrt{1.5\%} \\ &= 0.4\%\end{aligned}$$

C is incorrect. The covariance is not calculated solely from the standard deviations of the two assets. It also requires one use their correlation.

$$\begin{aligned}&= \sigma_1 \times \sigma_2 \\ &= \sqrt{\text{var}_1} \times \sqrt{\text{var}_2} \\ &= \sqrt{3.5\%} \times \sqrt{1.5\%} \\ &= 2.3\%\end{aligned}$$

Portfolio Risk and Return: Part I  
LOS c  
Section 4.1.3

**85** Among other things, an organization's risk tolerance should *most likely* reflect its:

- A** perception of market stability.
- B** size.
- C** competitive position.

C is correct. An organization with a strong competitive position can recover from losses more easily than one with a weaker competitive position. Therefore an organization's risk tolerance should reflect its competitive position. An organization's size does not define the risk sources it faces or the relative losses it can absorb, so it should not be reflected in its risk tolerance. Neither the risk sources affecting an organization nor the size of the losses an organization can absorb are a function of its perception of market stability.

A is incorrect because neither the risk sources affecting an organization nor the size of the losses an organization can absorb are a function of perceived market stability.

B is incorrect because an organization's size does not define the risk sources it faces or the relative losses it can absorb, so it should not be reflected in its risk tolerance.

Risk Management: An Introduction  
LOS d  
Section 3.2

- 86** ABC Fund invests in Singapore's government debt with maturities up to three months. It is *most likely* classified as a:
- A** fixed-income arbitrage fund.
  - B** money market fund.
  - C** bond mutual fund.

B is correct. Money market funds invest in short-term corporate or government debt. The difference between a bond mutual fund and a money market fund is the maturity of the underlying assets. In a money market fund, the maturity is as short as overnight and rarely longer than 90 days.

A is incorrect because a fixed-income arbitrage fund attempts to profit from arbitrage opportunities in interest rate securities.

C is incorrect because bond mutual funds hold bonds with maturities as short as one year and as long as 30 years.

Portfolio Management, An Overview  
LOS e  
Section 5.2.1

- 87** The economic equilibrium interest rate in a well-functioning financial system is *most likely* determined by:
- A** central bank policy.
  - B** the time consumption preferences of borrowers.
  - C** the supply and demand of money.

C is correct. The economic equilibrium interest rate in a well-functioning financial system is determined by the aggregate supply and demand of money. An expected rate of return exists, in theory, in which the aggregate supply of funds for investing and the aggregate demand for funds through borrowing and equity issuing are equal.

A is incorrect. Central bank policy is a specific example impacting the supply of money, but it does not address the demand of money.

B is incorrect. Time consumption preferences of borrowers are a specific example impacting the demand of money, but they do not address the supply of money.

Market Organization and Structure  
LOS a  
Sections 2.2, 4.2

- 88** An investor buys a stock on margin and holds the position for one year.

|                                |              |
|--------------------------------|--------------|
| Shares purchased               | 700          |
| Purchase price                 | \$22/share   |
| Call money rate                | 4%           |
| Dividend                       | \$0.60/share |
| Leverage ratio                 | 1.6          |
| Total return on the investment | 12%          |

Assuming that the interest on the loan and the dividend are both paid at the end of the year, the price at which the investor sold the stock is *closest* to:

- A \$23.05.
- B \$23.98.
- C \$23.38.

C is correct.

|                                                                                                  |                           |          |
|--------------------------------------------------------------------------------------------------|---------------------------|----------|
| Total purchase value = Purchase price × Shares purchased                                         | $\$22 \times 700$         | \$15,400 |
| Minus initial equity = Total purchase value/Leverage ratio                                       | $\$15,400/1.6$            | 9,625    |
| Amount borrowed = Total purchase value – Initial equity                                          | $\$15,400 - \$9,625$      | \$5,775  |
| Margin interest paid = Call money rate × Amount borrowed                                         | $4\% \times \$5,775$      | \$231    |
| Dividend income = Dividend per share × Shares purchased                                          | $\$0.60 \times 700$       | \$420    |
| Total return on the initial equity                                                               | $12\% \times \$9,625$     | \$1,155  |
| Gain from price appreciation = Total return – Dividend + Margin interest                         | $\$1,155 - \$420 + \$231$ | \$966    |
| Price at which investor sold the stock = Gain from price appreciation per share + Purchase price | $(\$966/700) + \$22$      | \$23.38  |

A is incorrect because it ignores the margin interest.

|                                                                                                  |                      |         |
|--------------------------------------------------------------------------------------------------|----------------------|---------|
| Gain from price appreciation = Total return – Dividend                                           | $\$1,155 - \$420$    | \$735   |
| Price at which investor sold the stock = Gain from price appreciation per share + Purchase price | $(\$735/700) + \$22$ | \$23.05 |

B is incorrect because it ignores the dividend amount.

|                                                                                                  |                        |         |
|--------------------------------------------------------------------------------------------------|------------------------|---------|
| Gain from price appreciation = Total return + Margin interest                                    | $\$1,155 + \$231$      | \$1,386 |
| Price at which investor sold the stock = Gain from price appreciation per share + Purchase price | $(\$1,386/700) + \$22$ | \$23.98 |

Market Organization and Structure  
LOS f  
Section 5.2

89 An equity fund manager is considering a market index as the benchmark for his portfolio, and he has the following preferences:

- the index should have a contrarian effect;
- shares held by controlling shareholders should be included;
- dividends should be included in the weighting of constituent securities; and
- the weights of constituent securities should not be arbitrarily determined by the index provider.

Which of the following weightings of indexes *best* meets the fund manager's preferences?

- A Equal
- B Float-adjusted market capitalization
- C Fundamental

C is correct. Fundamental weighting satisfies the fund manager's preferences. Fundamental indexes use a single measure, such as total dividends, to weight the constituent securities. Fundamentally weighted indexes generally will have a contrarian effect in that the portfolio weights will shift away from securities that have increased in relative value and toward securities that have fallen in relative value whenever the portfolio is rebalanced. All shares are included in a fundamental weighted index.

A is incorrect. The weights in an equal-weighted index are arbitrarily assigned by the index provider. The fund manager does not like this feature.

B is incorrect. In float-adjusted market-capitalization weighting, the shares held by controlling shareholders are excluded. The fund manager does not like this feature.

---

Security Market Indexes  
LOS d  
Section 3.2.4

- 90 A company that is required to raise equity capital to continue to operate as a going concern is *most likely* doing so to:
- A improve capital adequacy ratios.
  - B fund capital expansion projects.
  - C purchase long-lived assets.

A is correct. In cases in which a company must raise capital to ensure it can continue to operate as a going concern, capital is most likely raised to fulfill regulatory requirements, improve capital adequacy ratios, or ensure that debt covenants are met.

B is incorrect. A firm is not required to use equity capital to fund capital expansion projects. Funding capital expansion projects is a typical use of capital for companies that are raising capital to finance revenue generating activities, but this may not be the primary reason for a firm that is forced to raise capital to continue to operate as a going concern.

C is incorrect. A firm is not required to use equity capital to purchase long-lived assets. Purchasing long-lived assets is a typical use of capital for companies that are raising capital to finance revenue generating activities, but this may not be the primary reason for a firm that is forced to raise capital to continue to operate as a going concern.

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Overview of Equity Securities  
LOS f  
Section 7

- 91 The following data pertain to a company that can be appropriately valued using the Gordon growth model. The dividend is expected to grow indefinitely at the existing sustainable growth rate.

|                                      |        |
|--------------------------------------|--------|
| EPS growth rate (three-year average) | 7.50%  |
| Current dividend per share           | \$3.00 |
| Return on equity                     | 15%    |
| Dividend payout ratio                | 45%    |
| Investors' required rate of return   | 16%    |

The stock's intrinsic value is *closest* to:

- A \$34.62.
- B \$37.94.
- C \$41.90.

C is correct.

$$V_0 = \frac{D_0(1 + g)}{r - g}$$

where

$$\text{Sustainable growth rate} = g = b \times \text{ROE}$$

$$b = (1 - \text{Payout ratio})$$

$$g = (1 - 0.45) \times 15\% = 8.25\%$$

$$V_0 = (\$3 \times 1.0825) / (0.16 - 0.0825) = \$41.90$$

A is incorrect. It uses payout ratio instead of retention ratio ( $b$ ) to compute sustainable growth rate

$$g = 0.45 \times 15\% = 6.75\%$$

$$V_0 = \$3(1.0675) / (0.16 - 0.0675) = \$34.62$$

B is incorrect. It uses the EPS growth instead of sustainable growth rate.

$$V_0 = \$3(1.075) / (0.16 - 0.075) = \$37.94$$

Equity Valuation: Concepts and Basic Tools

LOS g

Section 4.2

92 Which of the following statements concerning different types of market indexes is *least* accurate?

- A Commodity indexes are subject to different weighting methods for specific commodities.
- B Equity indexes draw from a larger number of constituent securities than fixed-income indexes.
- C Hedge fund indexes are subject to survivorship bias from voluntary reporting.

B is correct. The fixed-income universe includes securities issued by governments, government agencies, and corporations with a variety of types and characteristics. The number of fixed-income securities is many times larger than the number of equity securities.



C is incorrect. Most research organizations rely on the voluntary cooperation of hedge funds to compile performance data. As a result, hedge fund indexes are subject to survivorship bias from voluntary reporting.

A is incorrect. Commodity indexes do not have an obvious weighting mechanism, and index providers create their own weighting methods. As a result, commodity indexes are subject to different weighting methods for specific commodities.

---

Security Market Indexes

LOS k

Sections 6.1, 7.1, 7.3

- 93 Which of the following is *most* accurate concerning key characteristics of different types of preference shares?
- A The price of convertible preference shares tends to be more volatile than their underlying common shares because they do not allow investors to share in profits of the company.
  - B Preference shareholders rank above subordinated bondholders with respect to claims on the company's net assets upon liquidation.
  - C Preference shares have characteristics of both debt and equity securities.



C is correct. Preference shares have characteristics of both debt securities and common shares. Similar to interest payments on debt securities, the dividends on preference shares are fixed but not contractual obligations. Similar to common shares, preference shares can be perpetual and can pay dividends indefinitely.

A is incorrect. The price of convertible preference shares is less volatile than their underlying common shares because there is less uncertainty about future cash flows as dividend payments are known and more stable. In addition, convertible preference shares allow investors the opportunity to share in the profits of the company.

B is incorrect. Both common and preference shareholders are owners of the company and will have claims on assets after liabilities to all creditors including bondholders have been completely paid. Preference shareholders rank before common shareholders but after bondholders to receive net distributions.

---

Overview of Equity Securities

LOS a, b, e

Section 3.2

- 94 An industry experiencing intense competitive rivalry among incumbent companies is *best* characterized by:
- A differentiated products and low exit barriers.
  - B a small number of competitors and low fixed costs.
  - C customers basing purchase decisions largely on price.



C is correct. The factor that most influences customer purchase decisions is likely to also be the focus of competitive rivalry in the industry. In general, industries in which price is a large factor in customer purchase decisions tend to be more competitive than industries in which customers value other attributes more highly.

A is incorrect. Industries experiencing more intense rivalry among incumbent companies are characterized by undifferentiated products and high exit barriers.

B is incorrect. Industries experiencing more intense rivalry among incumbent companies are fragmented among many small competitors and they tend to have high fixed costs.

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Introduction to Industry and Company Analysis  
LOS g, f  
Section 5.1.6

95 Unlike commercial industry classification systems, industry classification systems developed by governments *most likely*:

- A are updated more frequently.
- B are more transparent.
- C include private companies.

C is correct. Industry classification systems developed by governments do not distinguish between public and private companies, whereas commercial classification systems include only publicly traded organizations.

A is incorrect. Commercial industry classification systems are updated more frequently than government classification systems.

B is incorrect. Unlike commercial industry classification systems, most government classification systems do not disclose information about specific businesses.

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Introduction to Industry and Company Analysis  
LOS b  
Section 4.3

96 An investor uses the following data and Gordon's constant growth dividend discount model to evaluate a company's common stock. To estimate growth, she uses the average of the

- 1 average value of the compounded annual dividend growth rate over the period of 2006–2011, and
- 2 dividend payout ratio for the year 2011.

| Year | Earnings per Share | Dividend per Share | Return on Equity |
|------|--------------------|--------------------|------------------|
| 2011 | \$3.20             | \$1.92             | 12%              |
| 2010 | \$3.60             | \$1.85             | 17%              |
| 2009 | \$2.44             | \$1.74             | 13%              |
| 2008 | \$2.08             | \$1.62             | 15%              |
| 2007 | \$2.76             | \$1.35             | 11%              |
| 2006 | \$2.25             | \$1.25             | 9%               |

If her required return is 15%, the stock's intrinsic value is *closest* to:

- A \$30.14.
- B \$25.31.
- C \$23.71.

B is correct.  $V_0 = D_1/(r - g)$ ; First estimate the two growth rates.

- 1 Compound annual dividend growth rate over the period 2006–2011 =

$$1.25 \times (1 + g)^5 = 1.92$$

$$g = 8.96\% \approx 9\%$$

- 2 Sustainable growth rate for the year 2011 using the dividend payout ratio:

$$b = \text{earnings retention rate} = (1 - \text{Dividend payout ratio})$$

$$= [1 - (1.92/3.20)] = 0.40$$

$$g = b \times \text{ROE}; g = 0.40 \times 12\% = 4.8\%$$

$$\text{Average of the two approaches} = (9 + 4.8)/2 = 6.90\%$$

$$V_0 = D_1/(r - g)$$

$$= (1.92 \times 1.069)/(0.15 - 0.069)$$

$$= 2.05/0.081 = \$25.31$$

A is incorrect. It uses the payout ratio instead of the retention ratio in computing sustainable growth rate:  $g = 0.60 \times 12\% = 7.2\%$ ;

$$\text{Average of the two approaches} = (9 + 7.2)/2 = 8.1\%;$$

$$V_0 = D_1/(r - g) = \$1.92(1.081)/(0.15 - 0.081) = \$2.08/0.069 = \$30.14$$

C is incorrect. It uses  $D_0$  instead of  $D_1$ .

$$\$1.92/(0.15 - 0.069) = \$1.92/0.081 = \$23.71$$

Equity Valuation: Concepts and Basic Tools

LOS g

Section 4.2

- 97 Which of the following situations will *most likely* lead to a reduction of market efficiency?

- A An increase in rules and regulations that promote financial disclosure
- B An increase in arbitrage opportunities
- C An increase in the number of market participants

B is correct. Arbitrage is a set of transactions that produces riskless profits. Arbitrageurs are traders who engage in such trades to benefit from pricing discrepancies (inefficiencies) in markets. Such trading activity contributes to market efficiency. If arbitrage opportunities increase, it means that there are either more pricing discrepancies or fewer arbitrageurs (or both), and this situation will lead to a reduction in market efficiency.

A is incorrect. An increase of rules and regulations that promote financial disclosure will lead to an increase of market efficiency.

C is incorrect. An increase of the number of market participants will lead to an increase market efficiency.

Market Efficiency

LOS c

Section 2.3

- 98 A change in which of the following *best* describes a macroeconomic influence on industry growth?
- A The cost of debt
  - B Personal spending habits
  - C Population size



A is correct. External factors affecting an industry's growth include macroeconomic, technological, demographic, governmental, and social influences. A change in interest rates, or the cost of debt, is an example of a macroeconomic influence on industry growth, profitability, and risk.

C is incorrect. Changes in population size is an example of a demographic influence on industry growth, profitability, and risk.

B is incorrect. Changes in personal spending habits is an example of a social influence on industry growth, profitability, and risk.

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Introduction to Industry and Company Analysis  
LOS j  
Section 5.2

- 99 For bonds that are otherwise identical, the one exhibiting the highest level of positive convexity is *most likely* the one that is:
- A callable.
  - B option-free.
  - C puttable.



C is correct. When interest rates rise, a puttable bond is more likely to be put back to the issuer by the investor, limiting the loss of value and giving the bond more positive convexity than an option-free bond. In contrast, a callable bond is likely to be called from the investor when interest rates fall, limiting the gain in value and giving the bond negative convexity.

A is incorrect because a callable bond exhibits negative convexity.

B is incorrect because an option-free bond exhibits less positive convexity than a puttable bond.

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Understanding Fixed-Income Risk and Return  
LOS d  
Section 3.3

- 100 The market value of an 18-year zero-coupon bond with a maturity value of \$1,000 discounted at a 12% annual interest rate with semi-annual compounding is *closest* to:
- A \$130.04.
  - B \$192.86.
  - C \$122.74.

C is correct. The value of a zero-coupon bond is

$$\frac{\text{Face value}}{(1 + r)^N}$$

where  $r$  is the market discount rate per period, and  $N$  is the number of evenly spaced periods to maturity. The value of the zero-coupon bond is

$$\frac{\$1,000}{(1 + 0.12/2)^{18 \times 2}} = \$122.74$$

A is incorrect because it uses the annual discount rate and the maturity in years rather than adjusting the discount rate and maturity for semi-annual periods.

B is incorrect because it uses the semi-annual coupons times the number of years divided by 1.12.

---

Introduction to Fixed-Income Valuation  
LOS a  
Section 2.1

**101** Which of the following is *least likely* a component of the “Four Cs of Credit Analysis” framework?

- A** Covenants
- B** Competition
- C** Collateral

B is correct. The “Four Cs of Credit Analysis” framework includes capacity, collateral, covenants, and character. Competition is not one of the components.

A is incorrect because covenants are the terms and conditions of lending agreements that the issuer must comply with. It is part of the “Four Cs of Credit Analysis” framework.

C is incorrect because collateral refers to the quality and value of the assets supporting the issuer’s indebtedness. It is part of the “Four Cs of Credit Analysis” framework.

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Fundamentals of Credit Analysis  
LOS e  
Section 5.2

**102** The option-free bonds issued by ALS Corp. are currently priced at 108.50.

Based on a portfolio manager’s valuation model, a 1 bp increase in interest rates will result in the bond price falling to 108.40, whereas a 1 bp decrease in interest rates will result in the bond price rising to 108.59. The price value of a basis point (PVBp) for the bonds is *closest* to:

- A** 0.095.
- B** 0.088.
- C** 0.190.

A is correct. The bond's PVBP is computed using

$$\text{PVBP} = \frac{(PV_-) - (PV_+)}{2} = \frac{108.59 - 108.40}{2} = 0.095$$

B is incorrect because the PVBP is calculated using the incorrect formula:

$$\frac{108.59 - 108.40}{2 \times 108.50 \times 0.01} = 0.088$$

C is incorrect because the PVBP is calculated by omitting the 2 in the denominator:

$$108.59 - 108.40 = 0.190$$

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Understanding Fixed-Income Risk and Return  
LOS g  
Section 3.5

**103** To obtain the spot yield curve, a bond analyst would *most likely* use the most:

- A** recently issued and actively traded corporate bonds.
- B** recently issued and actively traded government bonds.
- C** seasoned and actively traded government bonds.

B is correct. To obtain the spot yield curve, a bond analyst would prefer to use the most recently issued and actively traded government bonds. Such bonds will have similar liquidity as well as fewer tax effects because they will be priced closer to par value.

A is incorrect because the spot yield curve is derived from the most recently issued and actively traded government, not corporate, bonds.

C is incorrect because seasoned government bonds tend to be less liquid than newly issued ones because they are typically owned by buy-and-hold investors.

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Introduction to Fixed-Income Valuation  
LOS g  
Section 4

**104** Which of the following terms in a bond issue *most likely* helps to reduce credit risk?

- A** Term maturity structure
- B** Floating-rate note
- C** Sinking fund arrangement

C is correct. A sinking fund arrangement is a way to reduce credit risk by making the issuer set aside funds over time to retire the bond issue.

A is incorrect because a term maturity structure is paid off in a lump sum at maturity and therefore carries more credit risk than a serial maturity structure such as sinking fund arrangement.

B is incorrect because a floating-rate note is a way to reduce interest rate risk instead of credit risk.

Fixed-Income Markets: Issuance, Trading, and Funding  
LOS g  
Section 6.3.3

- 105** The bonds of Whakatane and Co. are priced for settlement on 15 July 2014 and have the following features.

| Par value                | \$100.00               |
|--------------------------|------------------------|
| Annual coupon rate       | 8%                     |
| Coupon payment frequency | Semiannual             |
| Coupon payment dates     | 15 May and 15 November |
| Maturity date            | 15 November 2017       |
| Day count convention     | Actual/Actual          |
| Annual yield to maturity | 5.5%                   |

On the basis of this information, the *difference* between the full and flat prices is *closest* to:

- A 1.333.
- B 2.667.
- C 0.917.

A is correct. The difference between the full and flat prices is the accrued interest, which is computed as follows. Based on the Actual/Actual day convention, the number of days between the coupon periods is 183 days. Also, using the Actual/Actual day count convention, the number of days between 15 May 2014 and 15 July 2014 is 16 days remaining in May + 30 days in June + 15 days in July = 61 days. Accrued interest (per \$100 par value) =  $(61/183)(8.00/2) = 1.333$ .

B is incorrect because the accrued interest is computed using the annual coupon payment as: Accrued interest (per 100 par value) =  $(61/183)(8.00) = 2.667$ .

C is incorrect because the accrued interest is computed using the yield-to-maturity to compute the coupon payment as: Accrued interest (per 100 par value) =  $(61/183)(5.50/2) = 0.917$ .

Introduction to Fixed-Income Valuation  
LOS d  
Section 3.1

- 106** A bond portfolio manager is considering three bonds—A, B, and C—for his portfolio. Bond A allows the issuer to call the bond before the stated maturity, Bond B allows the investor to put the bond back to the issuer before the stated maturity, and Bond C contains no embedded options. The bonds are otherwise identical. The manager tells his assistant, “Bond A and Bond B should have larger nominal yield spreads to a US Treasury than Bond C to compensate for their embedded options.” Is the manager *most likely* correct?

- A No, Bond A’s nominal yield spread should be less than Bond C’s
- B No, Bond B’s nominal yield spread should be less than Bond C’s

**C** Yes

B is correct. Bond B's embedded put option benefits the investor, and the yield spread will therefore be less than the yield spread of Bond C, which does not contain this option or benefit.

A is incorrect because Bond A's embedded call option benefits the issuer, not the investor, therefore investors will demand a higher yield spread than on Bond C in compensation.

C is incorrect because Bond B should have a smaller yield spread.

Fixed-Income Securities: Defining Elements  
LOS f  
Section 5

**107** In primary bond markets, the method of allowing certain authorized issuers to offer additional bonds to the general public by preparing a single, all-encompassing offering circular is *most likely* known as a(n):

- A** private placement.
- B** shelf registration.
- C** underwritten offering.

B is correct. Under a shelf registration, the issuer prepares a single, all-encompassing offering circular that describes a range of future bond issuances, all under the same document. This master prospectus can be in place for years before it is replaced or updated, and it can be used to cover multiple bond issuances in the meantime.

A is incorrect because a private placement is a non-underwritten, unregistered offering of bonds that are sold only to an investor or a small group of investors.

C is incorrect because an underwritten offering guarantees the sale of the bond issue at an offering price that is negotiated with the issuer.

Fixed-Income Markets: Issuance, Trading and Funding  
LOS c  
Section 3.1

**108** Proceeds for repaying securitized bonds *most likely* come from the:

- A** claims-paying ability of the operating entity.
- B** cash flows of the project the bond is financing.
- C** cash flows of the underlying financial assets.

C is correct. Securitized bonds typically rely on the cash flows generated by one or more underlying financial assets as the primary source of the contractual payments to bondholders rather than on the claims-paying ability of the operating entity.

A is incorrect because the claims-paying ability of an operating entity is the source of payment for corporate bonds. These cash flows depend on the issuer's financial strength and integrity.



B is incorrect because the cash flows of the project the bond issue is financing are a major source for payment of non-sovereign government debt issues.

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Fixed-Income Securities: Defining Elements  
LOS b  
Section 3.1.2

**109** Which of the following is *least likely* a feature of a credit card receivable ABS?

- A An early amortization provision
- B Amortizing collateral
- C A lockout period



B is correct. A credit card receivable ABS is an example of an ABS with a non-amortizing collateral.

A is incorrect because a credit card receivable ABS may require early amortization of the principal if certain events occur. Such an early amortization provision would safeguard the credit quality of the issue.

C is incorrect because a credit card receivable ABS would typically have a lockout period during which the cash flow that is paid out to security holders is based only on finance charges collected and fees.

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Introduction to Asset-Backed Securities  
LOS h  
Sections 7.2

**110** Which investment will *most likely* expose investors to the greatest level of extension risk?

- A Shorter-term tranches in a collateralized mortgage obligation structure
- B Commercial mortgage-backed securities with a balloon payment
- C Planned amortization class tranches in a collateralized mortgage obligation structure



B is correct. Many commercial loans backing commercial mortgage-backed securities (CMBS) are balloon loans that require significant repayment of principal at maturity. The risk that the borrower will not be able to make the balloon payment is called balloon risk. The lender may decide to extend the loan over a period of time called the workout period. Because the term of the loan can be extended, balloon risk is a type of extension risk.

A is incorrect because shorter-term tranches in a CMO structure have less extension risk than longer-term tranches.

C is incorrect because planned amortization class tranches in a CMO structure offer investors reduction in both extension and contraction risk.

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Introduction to Asset-Backed Securities  
LOS f  
Section 5.2.1, 5.2.2, 6.2.2

**111** Convenience yield is *best* described as a nonmonetary benefit of holding a(n):

- A option contract.
- B asset.
- C forward contract.

B is correct. Convenience yield represents the nonmonetary advantage of holding the asset.

A is incorrect. Convenience yield is a benefit for the holder of the asset and not the holder of an option contract.

C is incorrect. Convenience yield is a benefit for the holder of the asset and not the holder of a forward contract.

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Basics of Derivative Pricing and Valuation  
LOS d  
Section 2.2.5

**112** According to put–call parity, if a fiduciary call expires in the money, the payoff is *most likely* equal to the:

- A difference between the market value of the asset and the face value of the risk-free bond.
- B market value of the asset.
- C face value of the risk-free bond.

B is correct. A fiduciary call, defined as a long position in a call and in a risk-free bond, generates a payoff that is equal to the market value of the asset if it expires in the money.

A is incorrect. The difference between the market value of the asset and the face value of the risk-free bond is the payoff of the long call if exercised. This ignores the fact that the face value of the bond needs to be added to the payoff.

C is incorrect. The face value of the risk-free bond is the payoff of the fiduciary call if the call expires out of the money,

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Basics of Derivative Pricing and Valuation  
LOS I  
Section 4.1.9

**113** In contrast to over-the-counter options, futures contracts *most likely*:

- A are not exposed to default risk.
- B represent a right rather than a commitment.
- C are private, customized transactions.

A is correct. Over-the-counter options are exposed to default risk, but futures contracts are standardized transactions that take place on futures exchanges and are not exposed to default risk.

B is incorrect. Futures contracts are commitments, but options represent a right.

C is incorrect. Futures contracts are not private; they are standardized instruments that trade on organized exchanges.

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Derivative Markets and Instruments  
LOS a, c  
Sections 2 and 3

**114** Over time, a forward contract *most likely* has variable:

- A** value and constant price.
- B** price and constant value.
- C** value and variable price.

A is correct. The price of a forward contract remains constant throughout its life. It is set as part of the contract specifications. The value varies with changes in the price of the underlying.

B is incorrect. The price is constant, but value varies with changes in the price of the underlying.

C is incorrect. The price is constant, but value varies with changes in the price of the underlying.

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Basics of Derivative Pricing and Valuation  
LOS b  
Section 2.4

**115** The price of an interest rate swap that involves the exchange of a fixed payment for a floating payment is *most likely*:

- A** equal to its value at expiration.
- B** set at initiation and constant over time.
- C** affected by changes in the floating payment.

B is correct. Swaps have both a price and a value. Price in the context of a swap is a reference to the fixed-rate payment on the swap, which is constant over time. The value of a swap is zero at initiation but can change over the life of the swap as market interest rates change.

A is incorrect. Price and value are not normally equal at expiration.

C is incorrect. The price in the context of a swap is a reference to the fixed-rate payment on the swap, which is constant over time and does not change in reaction to interest rate changes.

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Basics of Derivative Pricing and Valuation  
LOS h  
Section 3.3

**116** If an investor uses derivatives to make a long investment in commodities, the return earned on margin is *best* described as:

- A** convenience yield.
- B** collateral yield.

**C** price return.

B is correct. Collateral yield is the return on cash used as margin on derivatives used to gain commodity exposure.

A is incorrect because the convenience yield (also known as “roll yield”) is the return from rolling forward the maturity of the derivatives position.

C is incorrect because price return is the difference between the forward and spot price.

Introduction to Alternative Investments  
LOS e  
Section 6.4.1

**117** Which of the following investments *most likely* provides an investor with indirect equity exposure to real estate?

- A** Real estate limited partnerships
- B** Real estate investment trusts
- C** Commercial mortgage-backed securities

B is correct. Real estate investment trusts (REITs) provide investors with indirect equity real estate exposure. Real estate investment partnerships are a form of direct real estate equity investment. Commercial mortgage-backed securities (CMBS) provide investors with indirect debt investment opportunities in real estate.

A is incorrect. Real estate investment partnerships are a form of direct real estate equity investment.

C is incorrect. Commercial mortgage backed securities (CMBS) provide investors with indirect debt investment opportunities in real estate.

Introduction to Alternative Investments  
LOS d  
Sections 5.1, 5.2

**118** Investors look at many key due diligence factors when investing in hedge funds. Which of the following factors is *most likely* the biggest challenge to fully assess?

- A** Investment strategy and process
- B** Size and longevity
- C** Track record

A is correct. The investment strategy and process of a hedge fund is likely to be challenging to fully assess because hedge funds often limit disclosure in order to maintain their competitive advantage and to not give away information that is considered proprietary.

B is incorrect because the size and longevity of a hedge fund are common items for review and not as difficult to obtain as proprietary information.

C is incorrect because a hedge fund's track record should be readily available to investors.

---

Introduction to Alternative Investments  
LOS f  
Section 3.5

**119** The management fee of a private equity fund that has not yet invested all of its committed capital is *most likely* based on:

- A** committed capital.
- B** remaining capital.
- C** invested capital.



A is correct. The management fee of private equity funds is based on committed capital until the committed capital is fully drawn down and invested. This approach is in contrast to hedge funds, for which the management fee is based on invested capital.

B is incorrect. The management fee is based on committed capital and not remaining capital. Remaining capital serves as base for the fee once the committed capital is fully invested.

C is incorrect. The management fee is based on committed capital and not invested capital. This is an important distinction from hedge funds.

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Introduction to Alternative Investments  
LOS d  
Section 4.1

**120** Compared with traditional investments, over longer periods, alternative investments are *least likely* to have:

- A** better diversifying power.
- B** higher expected returns.
- C** more efficiently priced assets.



C is correct. Alternative investment strategies are more likely to include securities that trade in less liquid markets than securities that trade in relatively more liquid markets in which traditional, long-only investments trade.

A is incorrect because alternative investments generally have low return correlations with traditional asset markets, providing better diversification benefits.

B is incorrect because alternative investments are expected to produce higher returns (particularly on a risk-adjusted basis) than traditional markets, in part because they focus on less efficient markets.

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Introduction to Alternative Investments  
LOS a  
Section 2